

Immune Globulin (IVIG and SCIG)

ACG: A-0310 (AC)

[Link to Codes](#)

This MCG care guideline may overlap CMS guidance located in CMS statutes and regulations that are not currently part of MCG's Medicare Compliance content. For Medicare Advantage beneficiaries, before referring to MCG care guidelines, users should first follow all guidance related to this topic located in CMS National Coverage Determinations (NCDs), Local Coverage Determinations (LCDs), and other statutes and regulations. Content used in MCG care guidelines may be used to supplement but does not replace, modify, or supersede existing Medicare regulations or applicable NCDs or LCDs.

- Clinical Indications
- Evidence Summary
 - Background
 - Criteria
 - Inconclusive or Non-Supportive Evidence
 - Rationale
 - Related CMS Coverage Guidance
- References
- Footnotes
- Codes

Clinical Indications

- Immune globulin may be indicated for **1 or more** of the following(1)(2)(3)(4)(5):
 - ☐ Acquired red cell aplasia, as indicated by **1 or more** of the following[A](83)(84):[N](#)
 - Immunocompromised patient with aplasia associated with parvovirus B19 infection
 - Immunologic acquired red cell aplasia, with failure of or inability to tolerate first-line therapy (eg, prednisone, cyclophosphamide, cyclosporine)
 - Acute disseminated encephalomyelitis, with failure of treatment with corticosteroids, or corticosteroids contraindicated[B](30)[N](#)
 - ☐ Allogeneic hematopoietic stem cell transplant and **1 or more** of the following(87):[N](#)
 - Chronic graft vs host disease and recurrent sinopulmonary infection
 - Serum IgG less than 400 mg/dL (4 g/L) and age 18 years or younger
 - Serum IgG less than 400 mg/dL (4 g/L) and severe or recurrent infection (eg, sinopulmonary infection requiring hospitalization)
 - ☐ Anti-CD20 monoclonal antibody-related secondary hypogammaglobulinemia, as indicated by **ALL** of the following(88):[N](#)
 - History of recurrent or severe infection (eg, sinopulmonary infection requiring hospitalization)
 - Serum IgG 400 mg/dL (4 g/L) or less
 - Antiphospholipid syndrome, catastrophic[C](98)[N](#)
 - ☐ Autoimmune bullous disease, as indicated by **ALL** of the following[D](76)(101):[N](#)
 - Contraindications to, failure of, or significant side effects from systemic corticosteroids or immunosuppressive treatment
 - Dermatologic condition, as indicated by **1 or more** of the following:
 - Bullous pemphigoid(105)
 - Epidermolysis bullosa acquisita(105)(106)
 - Linear IgA bullous dermatosis
 - Mucous membrane (cicatricial) pemphigoid(103)(105)
 - Pemphigoid gestationis(105)
 - Pemphigus foliaceus
 - Pemphigus vulgaris(107)
 - Extensive and debilitating disease
 - Autoimmune encephalitis(16)(108)(109)(110)(111)[N](#)
 - Autoimmune hemolytic anemia, for severe life-threatening disease(115)[N](#)

- ☐ Chimeric antigen receptor (CAR) T-cell therapy-related toxicity, as indicated by **1 or more** of the following(118)(119)(120)(121)(122):[N](#)
 - Cytokine release syndrome, severe and refractory to high-dose corticosteroids and anti-interleukin-6 receptor antibody (anti-IL-6) (eg, tocilizumab)
 - Serum IgG less than 400 mg/dL (4 g/L) and age 18 years or younger
 - Serum IgG less than 400 mg/dL (4 g/L) and history of recurrent or severe infection (eg, sinopulmonary infection requiring hospitalization)
 - Chronic inflammatory demyelinating polyneuropathy(125)(126)(127)(129)(132)(133)(134)[N](#)
 - Dermatomyositis or polymyositis refractory to other immunosuppressive therapy or if immune globulin is being used as steroid-sparing option(108)(115)(139)(140)(141)[N](#)
- ☐ Fetal-neonatal alloimmune thrombocytopenia, as indicated by **1 or more** of the following(146)(147)(148):[N](#)
 - Newborn, if thrombocytopenia persists after transfusion of antigen-negative compatible platelets
 - Pregnant patient and **1 or more** of the following:
 - Family history of disease
 - Platelet alloantibodies found on screening
 - Previously affected pregnancy
- ☐ Guillain-Barre syndrome, as indicated by **ALL** of the following(30)(108)(150)(151)(152):[N](#)
 - Diagnosis of Guillain-Barre syndrome
 - Four weeks or less have elapsed since symptom onset.
 - Symptom evaluation, as indicated by **1 or more** of the following:
 - Patient able to walk only with assistance, or worse symptom severity
 - Progressive symptoms(157)
- ☐ Hematologic malignancy, as indicated by **1 or more** of the following:[N](#)
 - Acute lymphoblastic leukemia and serum IgG less than 400 mg/dL (4 g/L)(163)
 - Chronic lymphocytic leukemia and **1 or more** of the following(86)(159):
 - Corticosteroid-refractory autoimmune hemolytic anemia
 - History of recurrent or severe infection (eg, sinopulmonary infection requiring hospitalization), and serum IgG less than 500 mg/dL (5 g/L)
 - Multiple myeloma and **1 or more** of the following(158):
 - Patient receiving bispecific antibody (eg, elranatamab-bcmm, talquetamab-tgvs, teclistamab-cqyv), with serum IgG 400 mg/dL (4 g/L) or less at start of therapy(164)(165)
 - Patient receiving chimeric antigen receptor (CAR) T-cell therapy (eg, idecabtagene vicleucel, ciltacabtagene autoleucel) or within 1 year of starting CAR T-cell therapy, with serum IgG 400 mg/dL (4 g/L) or less at start of therapy(165)
 - Recurrent life-threatening infection
 - Serum IgG 400 mg/dL (4 g/L) or less
 - Non-Hodgkin lymphoma and **ALL** of the following(95)(166):
 - History of recurrent or severe infection (eg, sinopulmonary infection requiring hospitalization)
 - Serum IgG 400 mg/dL (4 g/L) or less
- ☐ Hemolytic disease of newborn, as indicated by **1 or more** of the following(167)(168)(169):[N](#)
 - Total serum bilirubin level within 2 mg/dL (34 micromoles/L) of age-adjusted and gestation-adjusted threshold for initiation of exchange transfusion
 - Total serum bilirubin still rising despite intensive phototherapy
- ☐ Hemolytic transfusion reaction and **1 or more** of the following(171)(172):[N](#)
 - After incompatible blood transfusion for severe life-threatening disease unresponsive to other therapies
 - Sickle cell disease and life-threatening post-transfusion hemolysis(173)
 - Hemophagocytic syndrome,^[I] for severe, persistent, or worsening disease(175)(176)(177)[N](#)
- ☐ HIV-positive status and **1 or more** of the following(179):[N](#)
 - Active bleeding and platelet count less than 10,000/mm³ (10 x10⁹/L)
 - Hypogammaglobulinemia and **ALL** of the following(180):
 - Age 18 years or younger
 - Serum IgG level less than 400 mg/dL (4 g/L)
 - Idiopathic (immune) thrombocytopenic purpura and need for rapid rise in platelet count to prevent or control bleeding(115)(125)(126)(127)(139)(159)(181)(182)(183)(184)(185)[N](#)
 - Immune checkpoint inhibitor-related toxicity^[K](118)(190)(191)(192)(193)(194)(195)(196)(197)[N](#)
- ☐ Infection prophylaxis, as indicated by **1 or more** of the following(12)(198)(199)(200):[N](#)
 - Hepatitis A infection prophylaxis,^[L] as indicated by **1 or more** of the following(198)(204):
 - Hepatitis A virus exposure within previous 14 days and **1 or more** of the following:
 - Age younger than 12 months or 40 years or older
 - Chronic liver disease
 - Hepatitis A vaccine contraindicated

- Immunocompromised patient
- Traveler to area with high or intermediate hepatitis A endemicity and **1 or more** of the following:
 - Age younger than 6 months or 40 years or older
 - Chronic liver disease
 - Hepatitis A vaccine contraindicated
 - Immunocompromised patient
- Measles infection prophylaxis,^[M] as indicated by **ALL** of the following(129)(199)(205):
 - Measles exposure within previous 6 days
 - Patient nonimmune to measles or with severe immunocompromise (eg, severe primary immunodeficiency, HIV infection with immunosuppression)
- Rubella infection prophylaxis,^[N] as indicated by **ALL** of the following(200):
 - Pregnant patient nonimmune to rubella
 - Rubella exposure within previous 3 days
- Varicella infection prophylaxis,^[O] as indicated by **ALL** of the following(202)(203)(206)(207)(208):
 - Patient at risk of varicella complications (eg, immunocompromised patient, neonate with maternal infection in perinatal period, premature neonate exposed postnatally, pregnant patient)
 - Patient nonimmune to varicella
 - Varicella exposure within previous 10 days
 - Varicella zoster-specific immune globulin unavailable
- Kawasaki disease^[P](159)(209)(210)(211)(212)(213)^[N]
- ☐ Kidney transplant and **1 or more** of the following(219)(220)(221):^[N]
 - Postoperative immune globulin needed for **ALL** of the following^[Q]:
 - Antibody-mediated transplant rejection
 - Planned plasmapheresis
 - Preoperative and perioperative immune globulin needed for **ALL** of the following^[R](219)(222):
 - Kidney transplant recipient has baseline anti-HLA antibody titer less than 1:16 to donor kidney.
 - Living donor transplant
- Lambert-Eaton syndrome, when steroids and other immunosuppressive treatments do not control symptoms(223)(224)(225)^[N]
- Multifocal motor neuropathy^[S](16)(108)(129)(229)(230)^[N]
- ☐ Multiple sclerosis, as indicated by **ALL** of the following(16)(30):^[N]
 - Clinical findings of multiple sclerosis, as indicated by **1 or more** of the following:
 - Relapsing-remitting multiple sclerosis
 - Severe refractory optic neuritis and **1 or more** of the following:
 - Corticosteroid therapy contraindicated
 - No vision recovery after 3 months of corticosteroid therapy
 - Treatment with standard immunomodulatory drugs (ie, interferon beta or glatiramer acetate) is not indicated due to **1 or more** of the following:
 - Failure of previous treatment
 - Intolerance to attempted previous treatment
 - Patient pregnant, post partum, or breast-feeding(108)(232)
 - Patient refusal
- Multisystem inflammatory syndrome in child in setting of coronavirus disease 2019 (COVID-19) infection^[T]^[U](233)(234)(235)(236)(237)^[N]
- ☐ Myasthenia gravis, as indicated by **ALL** of the following(16)(108)(151)(243)(244)(245):^[N]
 - Immune globulin not to be used for chronic maintenance therapy
 - Treatment of myasthenia gravis needed, as indicated by **1 or more** of the following:
 - Adult or juvenile myasthenia gravis and **1 or more** of the following(250):
 - Acute crisis
 - Acute exacerbation
 - Need for stabilization before surgery
 - Symptomatic patient resistant to or intolerant of immunosuppressive therapy
 - Neonatal myasthenia gravis
- Opsoclonus-myoclonus syndrome^[V](251)^[N]
- Post-transfusion purpura^[W](172)(255)^[N]
- ☐ Pregnancy-associated idiopathic (immune) thrombocytopenic purpura, as indicated by **1 or more** of the following(256)(257):^[N]
 - Any bleeding during pregnancy
 - Platelet count less than 30,000/mm³ (30 x10⁹/L) at any time during pregnancy
 - Platelet count less than level considered safe for invasive procedures (eg, cesarean birth, neuraxial anesthesia)
- ☐ Primary immunodeficiency, as indicated by **1 or more** of the following^[X](125)(126)(127)(129)(132)(159)(183)(184)(205)(259)(262)(265)(266)(267)(268):^[N]
 - Agammaglobulinemia (eg, less than 200 mg/dL (2 g/L))(267)

- Common variable immunodeficiency (CVID)(275)(276)
- Hyper-IgM syndrome (HIM)
- Primary hypogammaglobulinemia
- Serum IgG less than 400 mg/dL (4 g/L) and inadequate immunization response (ie, 4-fold increase in titers) to protein and polysaccharide antigens
- Rasmussen encephalitis, for short-term amelioration of symptoms(112)(277)(278) [N](#)
- Stevens-Johnson syndrome or toxic epidermal necrolysis, for life-threatening disease[Y](102)(280)(281) [N](#)
- Stiff-person syndrome, with failure of, or inability to receive or tolerate, GABA agonists (eg, benzodiazepine, baclofen)(16)(108) [N](#)
- Systemic lupus erythematosus, for disease unresponsive to other immunosuppressive therapies(51)(102)(285) [N](#)

Evidence Summary

Background

Immune globulin is prepared from the pooled plasma of several thousand donors to provide as complete a spectrum as possible of variable regions of antibody molecules found in the population.(1)(6)(7) **(EG 2)** The predominant component of immune globulin is IgG, distributed in the same proportion as in normal human serum; immune globulin also contains trace amounts of soluble CD4, HLA molecules, and cytokines.(6)(8) **(EG 2)** Modes of action include simple antibody replenishment as well as a variety of complex modulation processes involving several components of the immune response.(6)(7)(9) **(EG 2)** Several commercial preparations of immune globulin are available; however, there are insufficient data to recommend one preparation over another for use in treating a specific disease.(10) **(EG 2)** Immune globulin is not a generic drug, and immune globulin products should not be thought of as interchangeable; an immune globulin product needs to be selected in accordance with specific patient characteristics.(11) **(EG 2)** Immune globulin is available in intravenous (IVIG), subcutaneous (SCIG), and intramuscular forms.(1)(3)(12) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 215 published peer reviewed articles, 18 specialty society or other evidence-based guidelines, 26 Cochrane systematic reviews, and 1 book section.

For acquired red cell aplasia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** In a retrospective cohort study of 62 patients with pure red cell aplasia treated at a single institution, IVIG was associated with complete or partial response in all patients with parvovirus B19-associated pure red cell aplasia and pure red cell aplasia with hypogammaglobulinemia (4 and 11 patients treated with IVIG, respectively). IVIG was also used in 18 patients with idiopathic acquired pure red cell aplasia refractory to immunosuppressive therapy but was associated with a lower rate of response (22%).(84) **(EG 2)** Review articles state that IVIG is indicated for pure red cell aplasia associated with parvovirus B19 infection in immunocompromised patients on the basis of case series and case reports.(83)(85) **(EG 2)** An expert consensus guideline recommends IVIG for patients with chronic lymphocytic leukemia who develop pure red cell aplasia associated with parvovirus B19.(86) **(EG 2)** A review article notes that IVIG may be an appropriate option for patients with pure red cell aplasia who fail first-line therapy (eg, prednisone, cyclophosphamide, or cyclosporine).(83) **(EG 2)**

For acute disseminated encephalomyelitis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An expert consensus guideline suggests that IVIG may be effective in patients not responding to corticosteroids.(30) **(EG 2)**

For allogeneic hematopoietic stem cell transplant, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An international expert consensus guideline does not recommend routine use of IVIG in adults after hematopoietic stem cell transplant, but notes that it may be considered for patients with a serum IgG less than 400 mg/dL (4 g/L) and recurrent sinopulmonary infections and those with very low IgG levels (ie, less than 200 mg/dL (2 g/L)).(87) **(EG 2)** A specialty society guideline recommends against routine use of IVIG for infection prophylaxis after hematopoietic stem cell transplant but states that it may be considered for severe secondary hypogammaglobulinemia (ie, serum IgG less than 400 mg/dL (4 g/L)), particularly when associated with bacteremia or recurrent sinopulmonary infections despite antimicrobial prophylaxis.(88) **(EG 2)** Expert consensus guidelines recommend IVIG after allogeneic hematopoietic stem cell transplant for patients with severe hypogammaglobulinemia (ie, serum IgG less than 400 mg/dL (4 g/L)) and recurrent infections and in subsets of patients at high risk, including children undergoing transplant for disorders associated with B-cell deficiency and patients with chronic graft vs host disease with recurrent sinopulmonary infections.(89)(90) **(EG 2)** Another expert consensus guideline states that, while all patients undergoing allogeneic hematopoietic stem cell transplant should be considered as potential candidates for IVIG, particular consideration should be given to patients with IgG levels below 400 mg/dL (4 g/L) and those with graft vs host disease on immunosuppressive treatment.(91) **(EG 2)** An expert consensus guideline on supportive care during hematopoietic stem cell transplant in pediatric patients recommends IVIG for patients with serum IgG levels below 400 mg/dL (4 g/L).(92) **(EG 2)**

For secondary hypogammaglobulinemia related to anti-CD20 monoclonal antibody (eg, rituximab) therapy, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** B-cell targeted therapy with monoclonal antibodies directed against CD20, particularly rituximab, is associated

with a high risk of secondary hypogammaglobulinemia.(88) **(EG 2)** Specialty society and expert consensus guidelines recommend routine evaluation of serum immunoglobulin concentrations in rituximab-treated patients with autoimmune rheumatic diseases, with consideration for immunoglobulin replacement, particularly in patients with hypogammaglobulinemia and a history of serious, persistent, or recurrent infections.(80)(93)(94) **(EG 2)** Expert consensus guidelines on the management of lymphomas note that patients receiving anti-CD20 monoclonal antibody therapy are at risk for hypogammaglobulinemia and recommend consideration of IVIG in patients with frequent infections.(95)(96) **(EG 2)**

For catastrophic antiphospholipid syndrome, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** In an international registry study of 471 patients with catastrophic antiphospholipid syndrome, patients treated with anticoagulation, corticosteroids, and either plasma exchange or IVIG had a lower mortality rate than those who did not receive any of those therapies (28.6 % vs 75%) and those who received another combination of therapies or either anticoagulation or corticosteroids alone (28.6% vs 41.1%).(99) **(EG 2)** Based on a review of 6 studies (343 patients) evaluating IVIG in patients with catastrophic antiphospholipid syndrome that found a nonsignificantly lower mortality rate in IVIG-treated patients than non-IVIG-treated patients, an expert consensus guideline suggests the use of IVIG as first-line therapy for catastrophic antiphospholipid syndrome.(97) **(EG 2)** Another expert consensus guideline recommends IVIG as part of combination therapy with glucocorticoids and heparin for patients with catastrophic antiphospholipid syndrome.(100) **(EG 2)**

For autoimmune bullous diseases, including pemphigus vulgaris, pemphigus foliaceus, bullous pemphigoid, mucous membrane pemphigoid, epidermolysis bullous acquisita, pemphigoid gestationis, and linear IgA bullous dermatosis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert guidelines and reviews concluded that IVIG is indicated for patients refractory to corticosteroids or other immunosuppressive therapy.(76)(102) **(EG 2)** A systematic review of biologic treatment outcomes in 331 patients (154 of whom received IVIG) with mucous membrane pemphigoid found, at a median follow-up of 26.5 months, that IVIG treatment led to a complete response in 61.7% of patients, with a recurrence rate of 22.7% within 3.7 years of follow-up; the authors noted that the results were limited by the paucity of prospective studies and recommended future randomized trials.(103) **(EG 1)** A multicenter randomized controlled study of 56 patients with bullous pemphigoid that was resistant to prednisolone found that subsequent administration of IVIG was associated with a nonsignificant improvement, compared with placebo, in the primary endpoint of the disease activity score on day 15; the authors noted that the study may have been underpowered to detect differences in treatment groups with variable disease severity.(104) **(EG 1)**

For autoimmune encephalitis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Anti-N-methyl-D-aspartate receptor (NMDAR) encephalitis is characterized by a prodrome of headache and fever followed by sequential symptoms that may include behavioral changes, psychosis, catatonia, decreased level of consciousness, dyskinesias, and autonomic instability. Diagnosis is usually based on demonstration of NR1 IgG antibodies in serum or CSF, and initial therapy may include corticosteroids, IVIG, or plasma exchange.(109) (111)(112) **(EG 2)** A meta-analysis evaluating the use of immunotherapeutic agents in 1550 patients with NMDAR encephalitis (980 of whom received corticosteroids followed by IVIG as first-line therapy) found, at a median follow-up of 12 months, that use of corticosteroids plus IVIG, with or without apheresis, was associated with a good functional outcome (defined as a score of 0 to 2 on the Modified Rankin Scale).(113) **(EG 1)** Limbic encephalitis associated with voltage-gated potassium channel antibodies has been shown to respond within a few weeks to intense immunotherapy, which may include high-dose intravenous corticosteroids, IVIG, or plasma exchange.(109)(114) **(EG 2)** A review article states that IVIG can be given as first-line therapy for children with symptoms that suggest a moderate to high clinical suspicion of autoimmune encephalitis.(112) **(EG 2)**

For autoimmune hemolytic anemia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert opinion recommends IVIG for severe life-threatening disease but not for routine use in either acute or chronic disease.(116) **(EG 2)** In a case series of 56 patients with severe autoimmune hemolytic anemia, IVIG was associated with a clinical response in 67% of patients.(117) **(EG 2)**

For chimeric antigen receptor (CAR) T-cell therapy (eg, idecabtagene vicleucel, tisagenlecleucel)-related toxicity, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** CAR T-cell therapies directed against B-cell maturation antigen (eg, idecabtagene vicleucel) or CD19 (eg, axicabtagene ciloleucel, tisagenlecleucel) are frequently associated with B-cell aplasia and hypogammaglobulinemia, which may be prolonged.(118)(123)(124) **(EG 2)** An expert consensus guideline recommends IVIG for patients receiving CAR T-cell therapy with hypogammaglobulinemia (serum IgG less than 400 mg/dL (4 g/L) to 600 mg/dL (6 g/L)) who have serious or recurrent infections.(118) **(EG 2)** Another specialty society guideline recommends routine immunoglobulin replacement for pediatric patients receiving CAR T-cell therapy with immunoglobulin levels below age-adjusted normal ranges due to immunologic immaturity in childhood; for adults receiving CAR T-cell therapy, the authors recommend IVIG for hypogammaglobulinemia when there is a history of serious, recurrent, or chronic infections.(120) **(EG 2)** A specialty society guideline on the management of immune-related adverse events in patients receiving CAR T-cell therapy recommends IVIG for patients with IgG levels less than 400 mg/dL (4 g/L) and recurrent or life-threatening infections.(119) **(EG 2)** An expert consensus guideline notes that IVIG may also be considered for CAR T-cell therapy-related cytokine release syndrome that is refractory to corticosteroids and anti-interleukin-6 receptor antibody (anti-IL-6) therapy, but the available evidence is limited.(118) **(EG 2)**

For chronic inflammatory demyelinating polyneuropathy, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review articles and a specialty society guideline state that IVIG is an effective therapy.(128)(135)(136)(137) **(EG 2)** A

systematic review and meta-analysis of randomized trials comparing IVIG with other interventions for treatment of patients with definite or probable chronic inflammatory demyelinating polyneuropathy identified 5 trials comparing IVIG with placebo and found, at 2 to 6 weeks of follow-up, that IVIG was associated with improvement in disability compared with placebo, with persistent benefit seen up to 24 weeks of follow-up. Two trials comparing IVIG with corticosteroids (prednisolone or methylprednisolone) found, at 2 weeks of follow-up, no difference in disability improvement between groups.(133) **(EG 1)** A randomized phase III trial of 172 adult patients with chronic inflammatory demyelinating polyneuropathy who had previously responded to IVIG found that administration of low-dose or high-dose weekly SCIG for 24 weeks, compared with placebo, was effective at maintaining significant freedom from relapse in both active treatment groups; relapse rates were 63%, 39%, and 33%, respectively, in the placebo, low-dose active treatment, and high-dose active treatment groups, with low rates of adverse effects. The authors noted that this study did not directly compare IVIG with SCIG; in addition, data were missing on patients who withdrew from the study.(131) **(EG 1)** In a multicenter single-arm open-label trial, 49 patients with chronic inflammatory demyelinating polyneuropathy (typical or atypical) were treated with induction IVIG for 5 days followed by maintenance IVIG every 3 weeks; patients with a response at week 28 continued maintenance for a total of 52 weeks. At week 28, the responder rate was 77.6%; of the 34 patients who were enrolled in maintenance through the conclusion of the trial, 69.4% maintained improvement and 10.5% had a relapse.(138) **(EG 2)** An expert consensus guideline recommends IVIG for the treatment of chronic inflammatory demyelinating polyneuropathy.(4) **(EG 2)**

For dermatomyositis or polymyositis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Expert guidelines and review articles recommend IVIG for patients refractory to immunosuppressive therapy.(30)(76)(141)(142)(143) **(EG 2)** A systematic review found one randomized placebo-controlled trial, which revealed that treatment with IVIG produced statistically significant improvement in muscle strength over a course of 3 months.(140) **(EG 1)** A randomized controlled trial of 95 patients with active dermatomyositis compared treatment with either IVIG or placebo and found, at 16-week follow-up, that IVIG was associated with greater rates of disease improvement (defined as a composite Total Improvement Score of at least 20 and no deterioration while receiving treatment) compared with placebo; the authors recommended larger studies with longer follow-up to fully assess long-term efficacy and safety.(144) **(EG 1)** A review article found only 2 additional randomized controlled studies supporting the efficacy of IVIG in relapsed, refractory, or steroid-dependent disease, or as part of first-line therapy in older patients; the authors stressed the need for more robust evidence, using validated outcome measures to guide therapy decisions.(145) **(EG 1)** An expert consensus guideline recommends IVIG for the treatment of dermatomyositis but found insufficient evidence to recommend its use for polymyositis.(4) **(EG 2)**

For fetal-neonatal alloimmune thrombocytopenia, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review reported that IVIG, with or without corticosteroids, may be used as first-line therapy of fetal-neonatal alloimmune thrombocytopenia; however, the optimal timing and dosing have not been established.(149) **(EG 1)** A retrospective case series of 22 neonates suggests that antenatal weekly maternal IVIG is effective and safe for the treatment of fetal-neonatal alloimmune thrombocytopenia.(146) **(EG 2)** An expert consensus guideline considers IVIG to be standard first-line antenatal therapy as well as appropriate adjunctive therapy of a newborn, after first-line therapy with antigen-negative compatible platelets.(147) **(EG 2)**

For Guillain-Barre syndrome, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review and meta-analysis found comparable efficacy and safety for plasma exchange and IVIG for the management of Guillain-Barre syndrome.(153) **(EG 1)** A systematic review and meta-analysis of randomized controlled trials has confirmed effectiveness of IVIG as first-line therapy comparable to that of therapeutic plasma exchange when started within 2 weeks of onset of severe disease.(154) **(EG 1)** A randomized controlled trial of 93 patients with Guillain-Barre syndrome with poor prognosis (defined as a score of 6 or greater on the modified Erasmus Guillain-Barre Syndrome Outcome Score) treated with an initial dose of IVIG followed by either a second IVIG dose or placebo found, at 4-week follow-up, that a second IVIG dose was not associated with improved Guillain-Barre syndrome disability scores and resulted in more serious adverse events compared with placebo.(155) **(EG 1)** An expert consensus guideline on the diagnosis and treatment of Guillain-Barre syndrome in children and adolescents recommends IVIG for patients with severe or progressive disease.(156) **(EG 2)** Expert consensus guidelines recommend IVIG for the treatment of Guillain-Barre syndrome in adult and pediatric patients.(4)(152) **(EG 2)**

For hematologic malignancies, such as chronic lymphocytic leukemia or multiple myeloma, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert consensus guidelines recommend the use of IVIG for selected patients with recurrent or severe infections and low serum IgG levels, as well as for patients with chronic lymphocytic leukemia and corticosteroid-refractory autoimmune hemolytic anemia.(86)(158) (159) **(EG 2)** IVIG is recommended for patients with acute lymphoblastic leukemia who have serum IgG levels less than 400 mg/dL (4 g/L) and are considered at high risk for chemotherapy-related morbidity and mortality, particularly infants and patients with Down syndrome, or are undergoing CD19-targeted chimeric antigen receptor (CAR) T-cell therapy.(160)(161) **(EG 2)** An expert consensus guideline recommends that immunoglobulin replacement therapy be considered for patients with hematologic malignancies who have severe, recurrent, or persistent infections despite anti-infective therapy and have IgG levels less than 400 mg/dL (4 g/L) or fail a test immunization.(91) **(EG 2)** A systematic review of 21 randomized controlled trials evaluating interventions to reduce infections in patients with hematologic malignancies, including 8 studies comparing prophylactic immunoglobulin with placebo or no intervention and one study comparing immunoglobulin with antibiotics, found that prophylactic immunoglobulin was associated with a reduced risk of clinically documented infection, as compared with placebo or no intervention, but not associated with reduced risk of all-cause or infection-related mortality.(162) **(EG 1)**

For hemolytic disease of the newborn, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis found that

those studies suggesting benefit of IVIG for Rh hemolytic disease had the highest risks of bias, and efficacy of IVIG for treatment of ABO hemolytic disease was not clear for the same reason.(168) **(EG 1)** A systematic review and meta-analysis of 9 studies with 658 infants found that, while IVIG administration was associated with significantly reduced need for exchange transfusion, the quality of the evidence in general was low to very low, and the 2 studies with the lowest risk of bias showed no benefit of IVIG in reducing the need for or number of exchange transfusions.(170) **(EG 1)** Long-term follow-up of 66 patients with hemolytic disease of the newborn found that, after 2 to 7 years, IVIG administration, as compared with placebo, was associated with no significant difference in neurodevelopmental outcomes; the authors recommended further long-term studies.(169) **(EG 2)** A specialty society guideline recommends administration of IVIG in infants with isoimmune hemolytic disease and rising total serum bilirubin level.(167) **(EG 2)**

For hemolytic transfusion reaction, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** On the basis of case reports, review articles suggest the use of IVIG for acute hemolytic transfusion reactions, particularly for severe disease.(171)(172) **(EG 2)** A specialty society guideline on transfusion support for sickle cell disease suggests immunosuppressive therapy, including IVIG, for treatment of severe hemolytic transfusion reactions with hyperhemolysis.(173) **(EG 2)**

For hemophagocytic syndrome, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of 36 studies (493 patients) evaluating treatment of hemophagocytic lymphohistiocytosis in children (mean age 49.5 months) in the intensive care unit (161 of whom received IVIG) found that IVIG was associated with decreased mortality; the authors noted that the retrospective nature of some of the included studies limited the results, and further large prospective studies were recommended.(178) **(EG 1)** A specialty society guideline on the management of hemophagocytic lymphohistiocytosis in adults recommends the use of IVIG for clinically unstable or worsening patients with hemophagocytic syndrome secondary to a variety of causes (eg, infection, malignancy). (175) **(EG 2)** Another specialty society guideline recommends IVIG as a treatment option for suspected hemophagocytic lymphohistiocytosis in adults when there is persistent, severe, or worsening inflammation or organ dysfunction.(177) **(EG 2)**

For HIV-associated hematologic complications, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review suggests that IVIG is effective for rapidly increasing platelet count in patients who are actively bleeding.(179) **(EG 1)** For HIV-infected children, a systematic review and a practice guideline recommend IVIG for prevention of bacterial infection in children with coexistent hypogammaglobulinemia.(179)(180) **(EG 2)**

For idiopathic (immune) thrombocytopenic purpura, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review of randomized controlled trials reported that low-dose IVIG is effective in rapidly increasing the platelet count to prevent or control bleeding.(186) **(EG 1)** A systematic review and meta-analysis of 11 randomized trials including 558 children with immune thrombocytopenia compared treatment with IVIG or anti-D immunoglobulins and found, at 24 to 72 hours after administration, that IVIG was associated with more patients achieving a platelet count greater than 20,000/mm³ (20 x10⁹/L) compared with anti-D immunoglobulins.(187) **(EG 1)** A randomized trial of 206 pediatric patients (age 3 months to 16 years) with newly diagnosed idiopathic immune thrombocytopenia compared treatment with a single infusion of IVIG or observation and found, at 3-month follow-up, that IVIG was associated with more patients achieving a complete response (defined as a platelet count greater than 100,000/mm³ (100 x10⁹/L)) compared with observation; no difference in the development of chronic idiopathic immune thrombocytopenia or complete response rates at 6-month or 12-month follow-up was seen between the groups.(188) **(EG 1)** An expert consensus guideline recommends IVIG combined with corticosteroids in patients with idiopathic thrombocytopenic purpura when a rapid rise in platelets is warranted and suggests either IVIG or anti-D immunoglobulin in patients who require treatment and have contraindications to corticosteroids.(182) **(EG 2)** Another expert consensus guideline recommends IVIG as first-line treatment for children with acute immune thrombocytopenic purpura who require treatment (eg, for severe spontaneous mucosal bleeding).(189) **(EG 2)**

For immune checkpoint inhibitor-related toxicity, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Immune checkpoint inhibitors (eg, pembrolizumab, atezolizumab, nivolumab, ipilimumab) have been associated with a range of early-onset and late-onset inflammatory and autoimmune reactions that have typically been managed with immunosuppressive therapy.(118)(192) **(EG 2)** An expert consensus guideline recommends IVIG for management of immune checkpoint inhibitor-related myocarditis, demyelinating disease, and pneumonitis that is refractory to high-dose corticosteroids; severe immune checkpoint inhibitor-related bullous dermatitis and Stevens-Johnson syndrome/toxic epidermal necrolysis, myasthenia gravis, and Guillain-Barre syndrome; immune checkpoint inhibitor-related myositis that is severe, associated with dysphagia, or refractory to high-dose corticosteroids; and severe or rapidly progressing immune checkpoint inhibitor-related encephalitis.(118) **(EG 2)** A specialty society guideline on the management of immunotherapy-related toxicities recommends IVIG for grade 3 and above immune-related interstitial lung disease and corticosteroid-refractory pneumonitis, grade 3 and above immune-related myositis, and grade 3 and above immune-related neurologic symptoms.(192) **(EG 2)** Another specialty society guideline on immune checkpoint inhibitor-related adverse events recommends IVIG for grade 3 and above bullous dermatoses, grade 4 severe cutaneous adverse reactions, grade 3 and above corticosteroid-refractory pneumonitis, grade 3 and above myositis, grade 3 and above myasthenia gravis, Guillain-Barre syndrome of any severity, severe or progressing encephalitis, grade 3 and above demyelinating disease, grade 4 hemolytic anemia, and immune thrombocytopenic purpura, particularly when a rapid rise in platelets is needed.(190) **(EG 2)**

For infection prophylaxis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert consensus guidelines indicate that intramuscular immune globulin is indicated after hepatitis A exposure in individuals in whom hepatitis A vaccination is contraindicated, immunocompromised

patients, and those age 40 years or older or with chronic liver disease. Intramuscular immune globulin may also be indicated for individuals traveling to areas of high or intermediate hepatitis A endemicity for those age younger than 6 months or 40 years or older, who are unvaccinated, are immunocompromised, or with chronic liver disease.(198) **(EG 2)** Expert consensus guidelines recommend immune globulin for individuals at risk of measles (unvaccinated and no prior measles disease) after a known measles exposure.(199) **(EG 2)** Risk of rubella infection in nonimmune individuals may be reduced by immune globulin administration within 5 days of exposure.(201) **(EG 1)** Although there is limited evidence about the effectiveness of immune globulin for prevention of congenital rubella syndrome, expert opinion suggests that intramuscular immune globulin may be considered for pregnant patients who are nonimmune to rubella after a known rubella exposure.(200) **(EG 2)** For prevention of varicella infection, the use of varicella zoster-specific immune globulin is recommended for those without evidence of immunity and at high risk for complications (eg, immunocompromised or pregnant patients). However, IVIG may be administered when varicella zoster-specific immune globulin is unavailable.(202)(203) **(EG 2)**

For Kawasaki disease, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** IVIG is effective in reducing the incidence of coronary artery aneurysms as well as reducing fever and other clinical signs and symptoms.(211)(212)(214) (215)(216) **(EG 2)** A systematic review of 31 randomized controlled trials evaluating the efficacy of IVIG for the treatment of Kawasaki disease included 11 studies (1437 patients) comparing IVIG with aspirin and found moderate-certainty evidence that IVIG was associated with shorter fever duration and reduced incidence of coronary artery abnormalities at 30-day follow-up compared with aspirin.(217) **(EG 1)** An open-label randomized trial of 103 patients (age 4 weeks to 17 years) with Kawasaki disease resistant to IVIG (continued fever at least 36 hours after initial IVIG dose) compared treatment with either a second dose of IVIG or infliximab and found that infliximab was associated with more patients achieving resolution of fever, fewer total days of fever, and shorter hospital stay compared with IVIG, with no difference in coronary artery outcomes seen between groups. The authors recommended larger studies to further evaluate efficacy for coronary artery outcomes.(218) **(EG 1)**

For kidney transplant, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** An expert guideline recommends preoperative and perioperative IVIG for recipients with anti-HLA antibody sensitization to living donor kidneys.(72) **(EG 2)** A subsequent matched case control study demonstrated that preoperative treatment with IVIG and plasmapheresis provided a significant survival benefit for kidney transplant recipients with HLA antibody sensitization to living donor kidneys, as compared with patients on a transplant waiting list who had similar levels of panel-reactive antibodies.(219) **(EG 2)** For antibody-mediated rejection of a kidney transplant from either a deceased or a live donor, an expert consensus guideline recommends that IVIG be administered after each plasmapheresis treatment.(72) **(EG 2)**

For Lambert-Eaton syndrome, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review found only one small randomized crossover trial, which demonstrated effectiveness of IVIG in improving muscle strength for severe cases when steroids and other immunosuppressive treatments did not control symptoms; however, improvement lasted only for a period of several weeks.(226) **(EG 1)** A systematic review and meta-analysis of therapies for paraneoplastic neuropathies, including Lambert-Eaton syndrome, found no randomized controlled trials and only 6 studies involving 54 patients that suggested that IVIG is helpful.(227) **(EG 1)** An expert consensus guideline recommends IVIG for the treatment of Lambert-Eaton syndrome.(4) **(EG 2)**

For multifocal motor neuropathy, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review of 6 randomized trials (90 patients) evaluating the efficacy of immunoglobulin for the treatment of multifocal motor neuropathy found that IVIG treatment improved muscle strength and may improve disability; the authors noted that the small number of included patients limited the conclusions regarding disability improvement, and further randomized trials were recommended.(230) **(EG 1)** An expert consensus guideline recommends IVIG for the treatment of multifocal motor neuropathy.(4) **(EG 2)**

For multiple sclerosis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A specialty society guideline suggests that IVIG may have utility in the second-line or third-line treatment setting for relapsing-remitting multiple sclerosis in patients who do not tolerate or have contraindications to first-line therapies.(30) **(EG 1)** Evidence for effectiveness of IVIG for this condition is strongest in studies of the adult population; studies in the pediatric population are limited to a small number of case reports, with no reports as yet on the use of IVIG as a first-line therapy in pediatric multiple sclerosis.(231) **(EG 2)**

For multisystem inflammatory syndrome in children (MIS-C) in the setting of coronavirus disease 2019 (COVID-19) infection, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A multicenter, open-label, randomized controlled trial using an adaptive platform design randomized 214 inpatients with MIS-C to treatment with IVIG, methylprednisolone, or usual care and found that IVIG was not associated with an improved mean duration of hospital stay as compared with usual care (7.4 days in the IVIG group vs 7.6 days in the usual-care group). However, the authors noted that a high proportion of patients randomized to usual care received an off-protocol study treatment, including IVIG in 35% of patients assigned to usual care in the IVIG vs usual care comparison, which may have reduced the trial's ability to detect a treatment effect.(238) **(EG 1)** A multicenter, open-label, randomized controlled trial compared a 3-day course of intravenous methylprednisolone with a single dose of IVIG in 75 patients with MIS-C and found no between-group differences in the median hospital length of stay (6 days in the methylprednisolone group vs 6 days in the IVIG group). However, patients in the methylprednisolone group were less likely to require respiratory support than those in the IVIG group (27% vs 55%).(239) **(EG 1)** A multisite observational study of 518 patients (median age 8.7 years) with MIS-C evaluated the efficacy of immunomodulatory therapy (103 propensity score-matched patients received IVIG plus glucocorticoids and 103 received IVIG alone) and found that IVIG plus glucocorticoids was associated with lower risk of cardiovascular dysfunction (left ventricular dysfunction or shock resulting in use of

vasopressors) on or after day 2 of follow-up compared with IVIG alone.(240) **(EG 2)** An international observational study of 614 children with suspected MIS-C evaluated the efficacy of immunomodulatory therapy (246 received IVIG, 208 received IVIG plus glucocorticoids, and 90 received glucocorticoids alone) and found no difference between the 3 groups in disease severity reduction measured at day 2; the authors noted, however, that the conclusions were limited by the lack of randomization and the rapidly changing treatment environment, and they recommend larger studies with longer follow-up.(241) **(EG 2)** Expert consensus guidelines recommend IVIG as first-line treatment for MIS-C, in conjunction with corticosteroids with or without anakinra for patients with severe or refractory disease.(233)(234)(235)(236)(242) **(EG 2)**

For myasthenia gravis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review and a subsequent study concluded that IVIG may have effectiveness comparable to plasma exchange for temporary symptomatic improvement in severe exacerbations or acute crises.(244)(246)(247) **(EG 1)** A review article concluded that IVIG and plasma exchange are indeed considered equally effective in treating severe cases of myasthenia gravis.(243) **(EG 2)** A randomized controlled trial of 60 patients with generalized myasthenia gravis (all of whom were dependent on corticosteroids) compared treatment with either IVIG (loading dose followed by maintenance doses) or placebo and found, at 39-week follow-up, no difference between groups in the proportion of patients achieving a 50% or greater reduction in their corticosteroid dose.(248) **(EG 1)** A retrospective chart review of 52 patients treated with IVIG for myasthenia gravis found that 23 and 14 patients, respectively, achieved mild and moderate symptomatic improvement over an average of 5.9 years, but none achieved complete remission.(249) **(EG 2)** Guidelines from an international task force on myasthenia gravis conclude that IVIG is appropriate for short-term treatment for managing acute severe symptoms, in preparation for surgery, when rapid response to treatment is needed, when other therapies are ineffective, and before starting corticosteroids to minimize exacerbations. The authors note that chronic IVIG may be considered for patients with refractory myasthenia gravis.(245) **(EG 2)** An expert consensus guideline concludes that IVIG is effective for treating exacerbations of myasthenia gravis but does not recommend IVIG for the treatment of chronic stable disease.(4) **(EG 2)** Another expert consensus guideline recommends IVIG as an option for treating symptoms in active myasthenia gravis that is refractory to other therapies and for therapy in a myasthenic crisis. The authors recommend that IVIG be considered as maintenance therapy on an individualized basis, particularly in patients with a relative contraindication to other immunosuppressants and in patients with juvenile myasthenia gravis.(225) **(EG 2)**

For opsoclonus-myoclonus syndrome, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An open-label randomized trial of 53 patients (all age 8 years or younger) with neuroblastoma or ganglioneuroblastoma associated with opsoclonus-myoclonus syndrome compared treatment with prednisone combined with either cyclophosphamide or risk-adapted therapy with and without concurrent IVIG and found, at a median follow-up of 143 days, that IVIG was associated with greater improvement or normalization of opsoclonus-myoclonus-related symptoms after 1 year (81% and 41% in the IVIG and no-IVIG groups, respectively).(251) **(EG 1)** Expert opinion suggests that IVIG is a treatment option for opsoclonus-myoclonus syndrome secondary to both neoplastic and non-neoplastic causes.(253)(254) **(EG 2)**

For post-transfusion purpura, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert opinion suggests that IVIG is standard therapy for post-transfusion purpura.(172)(255) **(EG 2)**

For pregnancy-associated idiopathic (immune) thrombocytopenic purpura, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Specialty society guidelines recommend IVIG for treatment of immune thrombocytopenia during pregnancy.(256)(257)(258) **(EG 2)**

For primary immunodeficiency, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A specialty society guideline and reviews recommend IVIG as replacement therapy for patients with primary immunodeficiencies.(266)(269)(270) (271) **(EG 2)** A meta-analysis concluded that keeping trough IVIG levels at or above 1000 mg/dL (10 g/L) is likely associated with a significantly lower risk of pneumonia.(272) **(EG 1)** An observational study found that IVIG infusion resulted in significantly protective levels of various common antipneumococcal antibodies in patients with primary immunodeficiency.(273) **(EG 2)** A technology assessment found 16 studies suggesting that subcutaneous infusion of immunoglobulin for primary and secondary immunodeficiency may be as safe and effective as intravenous infusion of immunoglobulin, but the quality of the available evidence is low, significantly reducing the certainty of such findings.(274) **(EG 1)** For patients with specific antibody deficiency (ie, patients with recurrent respiratory tract infections, normal immunoglobulin and IgG subclass levels, and impaired response to pneumococcal capsular polysaccharide), a specialty society guideline notes that IgG replacement with IVIG may be indicated when other therapies (eg, repeat immunization with conjugate pneumococcal vaccine, antibiotics) have been ineffective.(266) **(EG 2)**

For Rasmussen encephalitis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles indicate, on the basis of case reports and case series, that IVIG may be useful for short-term therapy for Rasmussen encephalitis, a rare disorder characterized by focal epilepsy, lateralized brain atrophy, and cognitive impairment.(112)(277)(279) **(EG 2)** Smaller studies that are inconclusive suggest that IVIG may be effective as chronic therapy in patients who have not yet developed refractory epilepsy, with the caveat that the disease process may become arrested and prevent patients from reaching the point where they require surgery.(278) **(EG 2)**

For Stevens-Johnson syndrome or toxic epidermal necrolysis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review of 67 studies evaluating immunomodulating therapies for Stevens-Johnson syndrome/toxic epidermal necrolysis overlap or toxic epidermal

necrolysis identified 38 studies evaluating IVIG alone or in combination with other therapies (5 prospective comparative studies, 33 retrospective comparative studies or case series). In a network meta-analysis of 11 treatment arms, combination therapy with IVIG and corticosteroids was associated with improved mortality rates compared with mortality rates predicted by a toxic epidermal necrolysis-specific severity-of-illness score.(282) **(EG 1)** A meta-analysis of 13 observational studies found that higher dosages of IVIG correlated strongly with decreased mortality in patients with Stevens-Johnson syndrome and toxic epidermal necrolysis.(283) **(EG 1)** Expert consensus guidelines state that IVIG is an appropriate therapy for Stevens-Johnson syndrome or toxic epidermal necrolysis.(76)(102)(281) **(EG 2)**

For stiff-person syndrome, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Based on the results of a small randomized controlled trial, expert guidelines recommend IVIG for those failing first-line therapy with GABA agonist medications, such as baclofen or benzodiazepines.(30)(284) **(EG 2)** An expert consensus guideline concludes that IVIG is effective for treating stiff-person syndrome.(4) **(EG 2)**

For systemic lupus erythematosus, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert consensus guidelines and reviews suggest that IVIG may be appropriate in severely ill patients not responding to standard therapy,(76)(102)(286)(287) especially in those with concomitant infections.(285) **(EG 2)** A systematic review and meta-analysis found that the use of IVIG seems to be associated with significant reduction in disease activity and increase in complement levels, but the limited number of trials prevented assessment of other clinical outcome measures such as anti-double-stranded DNA, antinuclear antibody, steroid dosage, and renal function.(288) **(EG 1)** A retrospective case series describes complete remission after IVIG administration in 9 of 27 patients with active disease and concomitant infection, and in 6 of 26 patients with active disease resistant to standard therapy.(289) **(EG 2)** A specialty society guideline recommends IVIG as part of the treatment of severe autoimmune thrombocytopenia and catastrophic antiphospholipid syndrome in patients with systemic lupus erythematosus.(290) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For acute viral myocarditis, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review and meta-analysis identified 2 adult trials and one pediatric trial evaluating IVIG in acute presumed viral myocarditis. One placebo-controlled trial evaluated 62 adults with recent-onset dilated cardiomyopathy and found no significant difference between groups in terms of improvement in left ventricular ejection fraction or functional capacity. The second randomized trial of 41 adults with acute myocarditis without evidence of valvular or ischemic heart disease compared high-dose IVIG with no treatment and reported longer survival time in the IVIG-treated patients (at an average follow-up of 60 days); however, risk of bias and uncertain evidence quality limited the conclusions. A pediatric randomized trial evaluated the addition of IVIG to conventional therapy in 86 children with acute-onset dilated cardiomyopathy and found no difference between groups in overall survival, left ventricular end-diastolic diameter, or left ventricular shortening fraction at 6-month follow-up. The systematic review authors concluded that the evidence for IVIG for presumed viral myocarditis is uncertain and recommended additional higher-quality studies with low risk of bias and larger sample sizes.(13) **(EG 1)** A multi-institutional retrospective review of 216 pediatric patients with acute myocarditis found no evidence supporting the efficacy of IVIG in terms of improvement in cardiac function or survival; it stated that randomized controlled trials should be performed to determine if IVIG is appropriate for this condition.(14) **(EG 2)**

For Alzheimer disease, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A multicenter, randomized, placebo-controlled, phase II trial of 58 patients with mild to moderate Alzheimer disease found an effective safety and tolerability profile for IVIG, but this and other review articles cautioned that longer studies with more patients are required to assess possible effects on cognition and function.(15)(16)(17) **(EG 1)** A phase III double-blind placebo-controlled trial of 390 patients with mild to moderate Alzheimer disease found that biweekly administration of IVIG at 2 different dosages for 18 months, compared with placebo, resulted in no significant improvement in the outcome measures of cognition or improvement in activities of daily living; treatment with IVIG was associated with adverse effects including chills and rashes.(18) **(EG 1)**

For amyotrophic lateral sclerosis, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** An expert guideline states that there is insufficient evidence to recommend treatment with IVIG for amyotrophic lateral sclerosis.(19) **(EG 2)**

For atopic dermatitis, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review evaluating the efficacy of systemic treatments for atopic dermatitis included 3 studies (64 patients) of IVIG. One study in children reported improvements in atopic dermatitis severity with IVIG; other studies found no improvement with IVIG compared with placebo in adults or cyclosporine in children. The authors noted that all trials had high risk of bias and were limited by unclear reporting on patient age.(20) **(EG 1)**

For chronic fatigue syndrome, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** A systematic review of treatments for adults with myalgic encephalomyelitis or chronic fatigue syndrome found a single randomized controlled trial evaluating IVIG and concluded that there was no benefit from this therapy.(21) **(EG 1)** An expert guideline concluded that IVIG is unlikely to be beneficial and is not recommended for treatment of chronic fatigue syndrome.(22) **(EG 2)**

For complex regional pain syndrome, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A multicenter randomized placebo-controlled trial with 111 patients with moderate or severe complex regional pain syndrome and disease of 1 to 5 years' duration found no significant improvement in 24-hour average pain intensity or quality of life associated with 6 weeks of treatment with low-dose IVIG compared with placebo.(23) **(EG 1)**

For critical illness polyneuropathy, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A randomized controlled trial of 38 critically ill patients with multiple organ failure, systemic inflammatory response syndrome, and clinical signs of critical illness polyneuropathy with or without myopathy compared treatment with either IgM-enriched IVIG or placebo and found, at 14-day follow-up, no difference in symptoms severity scores between groups. The study was terminated early due to futility.(24) **(EG 1)** A review of neuromuscular complications of critical illness states that IVIG has not shown improvement in critical illness polyneuropathy.(25) **(EG 2)**

For dilated cardiomyopathy, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A pilot study of 17 patients with endomyocardial biopsy-proven high viral load of *Human parvovirus B19* found that administration of high-dose IVIG was associated with significant improvement in viral load, ejection fraction, and New York Heart Association functional class over a period of 12 months. However, the authors indicated that randomized placebo-controlled trials are needed.(26) **(EG 2)** A randomized controlled trial of 50 patients with chronic idiopathic dilated cardiomyopathy (all of whom were treated with standard medical therapy and had *Human parvovirus B19* DNA levels of greater than 200 copies/mcg on endomyocardial biopsy) compared treatment with either IVIG or placebo and found, at 6-month follow-up, that IVIG did not improve left ventricular ejection fraction or 6-minute walk test times compared with placebo.(27) **(EG 1)** In a randomized controlled study of 62 patients with acute myocardial infarction and decreased ejection fraction, administration of monthly IVIG for 6 months did not result in any significant improvement in either left ventricular remodeling or function.(28) **(EG 1)**

For epilepsy, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review found insufficient evidence to draw conclusions about the effectiveness of IVIG for epilepsy, and the authors stated that additional randomized controlled trials are needed.(29) **(EG 1)** An expert consensus guideline notes that IVIG seems to have a favorable effect in refractory epilepsy; however, this conclusion is based on limited evidence from case reports and small case series.(30) **(EG 2)**

For Graves ophthalmopathy, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** A systematic review and meta-analysis of randomized controlled trials found studies showing no incremental benefit of IVIG over standard therapy with oral corticosteroids and concluded that IVIG has marginal or unproven efficacy for this condition.(31) **(EG 1)**

For hematopoietic stem cell transplant without hypogammaglobulinemia, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** A systematic review and meta-analysis concluded that the use of IVIG as routine infection prophylaxis did not prevent infection or prolong survival.(32) **(EG 1)**

For IgG subclass deficiency with normal total IgG levels, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** An uncontrolled study of 10 patients with IgG subclass deficiency receiving monthly IVIG for 12 months reported significant improvement in infection rate, antibiotic use, and quality of life during IVIG administration. The authors indicated that larger randomized controlled studies will be required for confirmation.(33) **(EG 2)** A retrospective study of 13 patients treated with IVIG reported that 11 patients had improvement in the number of infections per year and 2 patients had no improvement. The authors stated that there was no correlation between response to IVIG and immunologic parameters, and the sample size was too small to reach a definitive conclusion.(34) **(EG 2)** An expert consensus guideline concluded that there is insufficient evidence to support adding IgG subclass deficiency or isolated IgA deficiency to the recommended indications for IVIG therapy.(35) **(EG 2)**

For inclusion body myositis, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** Expert guidelines state that there is insufficient evidence to support the use of IVIG for inclusion body myositis.(4)(30) **(EG 2)** A systematic review on interventions for dysphagia in long-term progressive muscle disease found a single randomized study involving patients with inclusion body myositis; because the study had a high risk of bias and did not specify some key outcomes, the authors concluded that there was insufficient evidence to determine if this intervention was efficacious for inclusion body myositis.(36) **(EG 1)**

For nephrotic syndrome, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review identified 4 studies that showed a beneficial effect of IVIG on the prevention of nosocomial infections in children with nephrotic syndrome; however, it was concluded that there is insufficient evidence to support the use of IVIG because the methodological quality of all studies was poor, the sample sizes were small, and all studies were from China.(37) **(EG 2)**

For neuromyelitis optica spectrum disorder, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A review article and a case series indicated that case-level evidence supports the use of IVIG in relapse treatment or prevention, especially in patients who cannot tolerate corticosteroids, but larger randomized trials are needed for confirmation.(16)(38)(39) **(EG 2)** A multicenter randomized controlled study initially targeting a total of 170 adults or children with acute first presentation of either transverse myelitis or neuromyelitis optica was to assess the effectiveness

of IVIG in addition to methylprednisolone alone during the first 5 days of steroid therapy, but the study was only able to recruit a total of 26 patients for screening, of which only 2 were randomized into the study. Consequently, there could be no determination of effectiveness of IVIG for these related conditions.(40) **(EG 1)** In some cases, optic neuritis may be a presentation of myelin oligodendrocyte glycoprotein antibody-associated disease (MOG-AD).(41)(42) **(EG 2)** A systematic review and network meta-analysis of 12 studies including 735 patients with MOG-AD (34 of whom received IVIG treatment) found that IVIG decreased the incidence of relapse compared with no treatment and that IVIG was the most effective of 10 treatments studied for reducing incidence of relapse. However, the authors noted that the retrospective nature of the included studies and the lack of standard dosing regimens limited the results, and larger prospective studies with standard regimens were recommended to verify the efficacy of IVIG.(43) **(EG 1)** Review articles note that IVIG has been used in case series and individual patients to treat neuromyelitis optica associated with myelin oligodendrocyte glycoprotein antibodies, but evidence for different treatment approaches is limited.(44)(45)(46)(47)(48) **(EG 2)**

For paraproteinemic demyelinating neuropathy, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review reported that there was inadequate evidence to support any immunomodulatory therapy, including IVIG, for this condition.(49) **(EG 2)** Expert guidelines have concluded that there is insufficient evidence to support the effectiveness of IVIG for this condition.(4)(50) **(EG 2)**

For pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANDAS), evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** Reviews of the medical literature have identified reports of IVIG having beneficial effects on tics and/or obsessive-compulsive disorder but have concluded that immunomodulator therapy such as IVIG has potential serious adverse effects and requires further research before it can be recommended.(51)(52) **(EG 2)**

For postpolio syndrome, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** A systematic review concluded that there is moderate-quality evidence that IVIG has no beneficial effect on activity limitations and that the evidence is inconsistent for effectiveness of IVIG on muscle strength and pain.(53) **(EG 1)** A randomized trial of 51 patients treated with either IVIG or placebo found that some components of health-related quality of life improved after 2 months, although gait, muscle strength, fatigue, and bodily pain did not change; overall, the primary endpoint was not met, and the authors indicated that further randomized studies are needed.(54) **(EG 1)** Expert guidelines have concluded that evidence is insufficient to support the use of IVIG for postpolio syndrome.(4)(30) **(EG 2)**

For the prevention of complete heart block (congenital), evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** Two open-label observational studies failed to demonstrate effectiveness of IVIG to prevent this condition in high-risk women who had previously delivered a child with complete heart block.(55)(56) **(EG 2)** A specialty society guideline on the diagnosis and treatment of fetal cardiac disease recommends in utero IVIG for the management of known fetal atrioventricular block but states that IVIG is not recommended for the prevention of heart block.(57) **(EG 2)**

For recurrent pregnancy loss or infertility, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review and meta-analysis evaluating the efficacy of immunotherapy on pregnancy outcomes in women with abnormal levels of natural killer cells and recurrent miscarriage or implantation failure included 4 nonrandomized trials of IVIG (557 patients) and found that treatment with IVIG improved live birth rates compared with control. However, the authors noted the results were limited by significant heterogeneity and lack of a standard dosing protocol, and further randomized studies to assess efficacy and safety were recommended.(58) **(EG 1)** A systematic review and meta-analysis identified 6 randomized trials studying the use of IVIG to improve the probability of a live birth in women with unexplained recurrent miscarriage; the authors found no benefit of IVIG for this indication and stated that it should not be used outside of randomized controlled trials.(59) **(EG 1)** A single-center, randomized, double-blind, placebo-controlled trial of 82 women with unexplained secondary recurrent miscarriages found no evidence that IVIG significantly increased the live birth rate, and the authors stated that it cannot be recommended in clinical practice.(60) **(EG 1)** A systematic review and meta-analysis identified 10 studies of patients with repeated in vitro fertilization failure and found that IVIG administration was associated with significantly improved implantation and clinical pregnancy rates, as well as a reduced rate of miscarriage; however, the incremental improvement in live birth rate per embryo transferred was not clinically significant, and the authors indicated that additional randomized controlled trials are needed.(61) **(EG 1)**

For rheumatic fever, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** A systematic review identified one study of the effect of using IVIG vs placebo to prevent cardiac disease in patients with acute rheumatic fever; the benefit was not statistically significant, and it was concluded that there is no evidence to support the use of IVIG to reduce the risk of developing heart valve lesions.(62) **(EG 1)**

For rheumatoid arthritis, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A review article states that while IVIG is used as an off-label treatment for various autoimmune diseases, effectiveness has not yet been confirmed for rheumatoid arthritis; several case series suggest, but do not confirm, efficacy in patients for whom standard therapies are contraindicated or intolerable, including those with Felty syndrome, Still disease, pregnancy with active disease, or infection in the setting of active disease.(63) **(EG 2)**

For sepsis or septic shock in adults, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** A systematic review and meta-analysis and a review article found some evidence that polyclonal IVIG reduced mortality in adults with

sepsis, but this benefit was not observed in trials with lower risks of bias; for IgM-enriched IVIG and monoclonal IVIG, evidence was found to be insufficient to make any conclusions.(2)(64) **(EG 1)**

For sepsis or septic shock in neonates, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** Systematic reviews and meta-analyses found sufficient evidence that treatment using monoclonal, polyclonal, or IgM-enriched IVIG does not reduce mortality in neonates with proven or suspected sepsis. (64)(65)(66) **(EG 1)** A multicenter randomized placebo-controlled trial of 3490 infants receiving antibiotics for known or suspected sepsis reported that the addition of IVIG had no effect on adverse events, disability, death, or subsequent septic episodes.(67) **(EG 1)** For preventing infection in preterm and/or low-birth-weight infants, a systematic review and meta-analysis of 19 studies of approximately 5000 infants concluded that IVIG was effective, with a significant absolute risk reduction of 3% in new cases of sepsis and 4% in one or more episodes of serious infection; however, no significant reductions were observed in other clinical outcomes, including mortality.(68) **(EG 1)**

For systemic sclerosis with diffuse cutaneous manifestations, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A multicenter randomized controlled study of 63 patients found no statistically significant differences between IVIG and placebo in terms of disease resolution after a single course of IVIG but found some evidence that a second course, given to nonresponders, may be effective. The authors cautioned that such findings are preliminary, and larger randomized studies are necessary.(69) **(EG 1)** A review article found only a small number of case series or clinical trials that suggested possible improvement in skin, joint, and interstitial lung symptoms for patients with systemic sclerosis and stressed that additional randomized trials are necessary to confirm both effectiveness and dosage.(70) **(EG 2)** An expert consensus guideline includes IVIG as a potential treatment option for skin involvement in juvenile systemic sclerosis when symptoms are refractory to first-line and second-line therapy.(71) **(EG 2)**

For transplant of solid organs other than the kidney, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** An expert guideline found insufficient evidence to support the use of IVIG either to prolong graft survival or to treat rejection for heart, lung, or liver transplant.(72) **(EG 2)** A review article found that evidence is beginning to build support for IVIG in this setting; however, conclusive information is not yet available.(73) **(EG 2)** A review of therapies for antibody-mediated rejection in cardiac transplant states that IVIG, often in combination with other agents, has never been systematically studied for the purpose of prophylactically reducing the incidence of rejection; however, small case series suggest possible efficacy.(74) **(EG 2)**

For urticaria that is refractory to standard therapy, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A review article indicates that multiple case reports suggest that IVIG may be useful in patients with chronic disease who are dependent on corticosteroids or cyclosporine, or who may need to discontinue these medications due to side effects; the authors indicate that larger confirmatory studies are necessary.(75) **(EG 2)** An expert consensus guideline found only case reports and small case series that suggested efficacy of IVIG for autoimmune urticaria; larger controlled studies are needed.(76) **(EG 2)**

For vasculitis due to antineutrophil cytoplasmic antibodies, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review and meta-analysis of 9 studies with 184 patients (including one randomized trial) evaluating IVIG for antineutrophil cytoplasmic antibody-associated vasculitis found, in pooled analysis, that IVIG was associated with improvements from baseline in vasculitis activity scores, C-reactive protein, and antineutrophil cytoplasmic antibody levels. However, the authors noted that the available evidence is limited by a lack of comparative studies and that randomized controlled trials are needed.(77) **(EG 1)** A meta-analysis evaluating treatment for granulomatosis with polyangiitis found insufficient evidence that IVIG provides a therapeutic advantage to a standard combination of steroids and immunosuppressants, suggesting that its use for this condition be limited to randomized controlled trials.(78) **(EG 1)** A randomized placebo-controlled trial of 37 patients with antineutrophil cytoplasmic antibody-associated microscopic polyangiitis with corticosteroid-refractory neuropathy compared treatment with either IVIG (2 doses given 4 weeks apart) or placebo and found, at 8-week follow-up, no significant between-group differences in change from baseline in manual muscle test score.(79) **(EG 1)** A specialty society practice guideline states that IVIG should not be used routinely to treat granulomatosis with polyangiitis or microscopic polyangiitis but notes that there may be instances when short-term IVIG may be considered (ie, refractory disease or contraindication to remission induction therapy).(80) **(EG 2)** An international consensus guideline on the management of antineutrophil cytoplasmic antibody-associated vasculitis does not include IVIG as a recommended therapy for induction of remission or maintenance therapy but notes that it may be considered in the setting of diffuse alveolar bleeding with hypoxemia.(81) **(EG 2)**

For viral encephalitis in children, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review and meta-analysis of 3 randomized controlled trials (138 patients) evaluating IVIG for treatment of viral encephalitis in pediatric patients found, for children with Japanese encephalitis, no difference in disability or serious adverse events between IVIG and placebo. While there was evidence of clinical benefit of adjunctive IVIG treatment for children with any encephalitis (eg, shorter time to regain consciousness and resolution of symptoms, shorter length of hospital stay), the evidence was deemed low-quality with high risk of bias; the authors could not make firm conclusions regarding the efficacy of this treatment, and further randomized trials were recommended.(82) **(EG 1)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(291); 42 CFR 422.101(292)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(293); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(294); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(295)

Medicare Coverage Determinations: Medicare Coverage Database(296)

References

1. Gelfand EW. Intravenous immune globulin in autoimmune and inflammatory diseases. *New England Journal of Medicine* 2012;367(21):2015-2025. DOI: 10.1056/NEJMra1009433. [Context Link 1, 2, 3] View abstract...
2. Wang J, McQuilten ZK, Wood EM, Aubron C. Intravenous immunoglobulin in critically ill adults: When and what is the evidence? *Journal of Critical Care* 2015;30(3):652.e9-e16. DOI: 10.1016/j.jcrc.2015.01.022. [Context Link 1, 2] View abstract...
3. Perez EE, et al. Update on the use of immunoglobulin in human disease: A review of evidence. *Journal of Allergy and Clinical Immunology* 2017;139(3S):S1-S46. DOI: 10.1016/j.jaci.2016.09.023. [Context Link 1, 2] View abstract...
4. Tavee J, et al. Updated consensus statement: Intravenous immunoglobulin in the treatment of neuromuscular disorders report of the AANEM ad hoc committee. *Muscle and Nerve* 2023;68(4):356-374. DOI: 10.1002/mus.27922. [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11] View abstract...
5. Grigoriadou S, et al. British Society for Immunology and United Kingdom Primary Immunodeficiency Network (UKPIN) consensus guideline for the management of immunoglobulin replacement therapy. *Clinical and Experimental Immunology* 2022;210(1):1-13. DOI: 10.1093/cei/uxac070. [Context Link 1] View abstract...
6. Schwab I, Nimmerjahn F. Intravenous immunoglobulin therapy: how does IgG modulate the immune system? *Nature Reviews. Immunology* 2013;13(3):176-189. DOI: 10.1038/nri3401. [Context Link 1, 2, 3] View abstract...
7. Quinti I, Coluzzi S, Pulvirenti F, Prezzo A, Girelli G. Polyvalent immunoglobulins: challenges and perspectives. *Blood Transfusion* 2013;11 Suppl 4:s40-s44. DOI: 10.2450/2013.008s. [Context Link 1, 2] View abstract...
8. Saeedian M, Randhawa I. Immunoglobulin replacement therapy: a twenty-year review and current update. *International Archives of Allergy and Immunology* 2014;164(2):151-166. DOI: 10.1159/000363445. [Context Link 1] View abstract...
9. Dalakas MC, Latov N, Kuitwaard K. Intravenous immunoglobulin in chronic inflammatory demyelinating polyradiculoneuropathy (CIDP): mechanisms of action and clinical and genetic considerations. *Expert Review of Neurotherapeutics* 2022;22(11-12):953-962. DOI: 10.1080/14737175.2022.2169134. [Context Link 1] View abstract...
10. Gurcan HM, Keskin DB, Ahmed AR. Information for healthcare providers on general features of IGIV with emphasis on differences between commercially available products. *Autoimmunity Reviews* 2010;9(8):553-559. DOI: 10.1016/j.autrev.2010.03.003. [Context Link 1] View abstract...
11. Eight Guiding Principles for Effective Use of IVIG for Patients With Primary Immunodeficiency. [Internet] American Academy of Allergy Asthma and Immunology. 2011 Dec Accessed at: <https://www.aaaai.org/>. [accessed 2024 Sep 24] [Context Link 1, 2]
12. GamaSTAN® (immune globulin (human)). Physician Prescribing Information [Internet] Grifols Therapeutics, Inc. 2022 Aug Accessed at: <https://www.gamastan.com/en/hcp>. [created 1944; accessed 2024 Sep25] [Context Link 1, 2]
13. Robinson J, Hartling L, Vandermeer B, Sebastianski M, Klassen TP. Intravenous immunoglobulin for presumed viral myocarditis in children and adults. *Cochrane Database of Systematic Reviews* 2020, Issue 8. Art. No.: CD004370. DOI: 10.1002/14651858.CD004370.pub4. [Context Link 1] View abstract...
14. Klugman D, Berger JT, Sable CA, He J, Khandelwal SG, Slonim AD. Pediatric patients hospitalized with myocarditis: a multi-institutional analysis. *Pediatric Cardiology* 2010;31(2):222-228. DOI: 10.1007/s00246-009-9589-9. [Context Link 1] View abstract...
15. Dodel R, et al. Intravenous immunoglobulin for treatment of mild-to-moderate Alzheimer's disease: a phase 2, randomised, double-blind, placebo-controlled, dose-finding trial. *Lancet Neurology* 2013;12(3):233-243. DOI: 10.1016/S1474-4422(13)70014-0. [Context Link 1] View abstract...
16. Lunemann JD, Nimmerjahn F, Dalakas MC. Intravenous immunoglobulin in neurology--mode of action and clinical efficacy. *Nature Reviews. Neurology* 2015;11(2):80-89. DOI: 10.1038/nrneurol.2014.253. [Context Link 1, 2, 3, 4, 5, 6, 7] View abstract...
17. Kile S, et al. IVIG treatment of mild cognitive impairment due to Alzheimer's disease: a randomised double-blinded exploratory study of the effect on brain atrophy, cognition and conversion to dementia. *Journal of Neurology, Neurosurgery, and Psychiatry* 2017;88(2):106-112. DOI: 10.1136/jnnp-2015-311486. [Context Link 1] View abstract...

18. Relkin NR, et al. A phase 3 trial of IV immunoglobulin for Alzheimer disease. *Neurology* 2017;88(18):1768-1775. DOI: 10.1212/WNL.0000000000003904. [Context Link 1] View abstract...
19. EFNS Task Force on Diagnosis and Management of Amyotrophic Lateral Sclerosis; et al. EFNS guidelines on the clinical management of amyotrophic lateral sclerosis (MALS)--revised report of an EFNS task force. *European Journal of Neurology* 2012;19(3):360-375. DOI: 10.1111/j.1468-1331.2011.03501.x. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
20. Siegels D, et al. Systemic treatments in the management of atopic dermatitis: A systematic review and meta-analysis. *Allergy* 2021;76(4):1053-1076. DOI: 10.1111/all.14631. [Context Link 1] View abstract...
21. Smith ME, et al. Treatment of myalgic encephalomyelitis/chronic fatigue syndrome: a systematic review for a National Institutes of Health Pathways to Prevention workshop. *Annals of Internal Medicine* 2015;162(12):841-850. DOI: 10.7326/M15-0114. [Context Link 1] View abstract...
22. Provan D, Chapel HM, Sewell WA, O'Shaughnessy D, UK Immunoglobulin Expert Working Group. Prescribing intravenous immunoglobulin: summary of Department of Health guidelines. *British Medical Journal* 2008;337:a1831. [Context Link 1] View abstract...
23. Goebel A, et al. Low-dose intravenous immunoglobulin treatment for long-standing complex regional pain syndrome: a randomized trial. *Annals of Internal Medicine* 2017;167(7):476-483. DOI: 10.7326/M17-0509. [Context Link 1] View abstract...
24. Brunner R, et al. Early treatment with IgM-enriched intravenous immunoglobulin does not mitigate critical illness polyneuropathy and/or myopathy in patients with multiple organ failure and SIRS/sepsis: a prospective, randomized, placebo-controlled, double-blinded trial. *Critical Care* 2013;17(5):R213. DOI: 10.1186/cc13028. [Context Link 1] View abstract...
25. Osias J, Manno E. Neuromuscular complications of critical illness. *Critical Care Clinics* 2014;30(4):785-94. DOI: 10.1016/j.ccc.2014.06.008. [Context Link 1] View abstract...
26. Dennert R, et al. Intravenous immunoglobulin therapy for patients with idiopathic cardiomyopathy and endomyocardial biopsy-proven high PVB19 viral load. *Antiviral Therapy* 2010;15(2):193-201. DOI: 10.3851/IMP1516. [Context Link 1] View abstract...
27. Hazebroek MR, et al. Intravenous immunoglobulin therapy in adult patients with idiopathic chronic cardiomyopathy and cardiac parvovirus B19 persistence: a prospective, double-blind, randomized, placebo-controlled clinical trial. *European Journal of Heart Failure* 2021;23(2):302-309. DOI: 10.1002/ejhf.2082. [Context Link 1] View abstract...
28. Gullestad L, et al. Intravenous immunoglobulin does not reduce left ventricular remodeling in patients with myocardial dysfunction during hospitalization after acute myocardial infarction. *International Journal of Cardiology* 2013;168(1):212-218. DOI: 10.1016/j.ijcard.2012.09.092. [Context Link 1] View abstract...
29. Geng J, et al. Intravenous immunoglobulins for epilepsy. *Cochrane Database of Systematic Reviews* 2019, Issue 12. Art. No.: CD008557. DOI: 10.1002/14651858.CD008557.pub4. [Context Link 1] View abstract...
30. Elovaara I, et al. EFNS guidelines for the use of intravenous immunoglobulin in treatment of neurological diseases: EFNS task force on the use of intravenous immunoglobulin in treatment of neurological diseases. *European Journal of Neurology* 2008;15(9):893-908. DOI: 10.1111/j.1468-1331.2008.02246.x. (Reaffirmed 2024 Aug) [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11] View abstract...
31. Stiebel-Kalish H, Robenshtok E, Hasanreisoglu M, Ezrachi D, Shimon I, Leibovici L. Treatment modalities for Graves' ophthalmopathy: systematic review and metaanalysis. *Journal of Clinical Endocrinology and Metabolism* 2009;94(8):2708-2716. DOI: 10.1210/jc.2009-0376. [Context Link 1] View abstract...
32. Raanani P, Gafter-Gvili A, Paul M, Ben-Bassat I, Leibovici L, Shpilberg O. Immunoglobulin prophylaxis in hematological malignancies and hematopoietic stem cell transplantation. *Cochrane Database of Systematic Reviews* 2008, (verified by Cochrane 2010 Jan), Issue 4. Art. No.: CD006501. DOI: 10.1002/14651858.CD006501.pub2. [Context Link 1] View abstract...
33. Abdou NI, Greenwell CA, Mehta R, Narra M, Hester JD, Halsey JF. Efficacy of intravenous gammaglobulin for immunoglobulin G subclass and/or antibody deficiency in adults. *International Archives of Allergy and Immunology* 2009;149(3):267-274. DOI: 10.1159/000199723. [Context Link 1] View abstract...
34. Abrahamian F, Agrawal S, Gupta S. Immunological and clinical profile of adult patients with selective immunoglobulin subclass deficiency: response to intravenous immunoglobulin therapy. *Clinical and Experimental Immunology* 2010;159(3):344-350. DOI: 10.1111/j.1365-2249.2009.04062.x. [Context Link 1] View abstract...
35. Delforge M, et al. Recommended indications for the administration of polyclonal immunoglobulin preparations. *Acta Clinica Belgica* 2011;66(5):346-360. [Context Link 1] View abstract...
36. Jones K, et al. Interventions for dysphagia in long-term, progressive muscle disease. *Cochrane Database of Systematic Reviews* 2016, Issue 2. Art. No.: CD004303. DOI: 10.1002/14651858.CD004303.pub4. [Context Link 1] View abstract...
37. Wu HM, Tang JL, Cao L, Sha ZH, Li Y. Interventions for preventing infection in nephrotic syndrome. *Cochrane Database of Systematic Reviews* 2012, Issue 4. Art. No.: CD003964. DOI: 10.1002/14651858.CD003964.pub3. [Context Link 1] View abstract...
38. Wingerchuk DM. Neuromyelitis optica: potential roles for intravenous immunoglobulin. *Journal of Clinical Immunology* 2013;33 Suppl 1:S33-S37. DOI: 10.1007/s10875-012-9796-7. [Context Link 1] View abstract...
39. Magraner MJ, Coret F, Casanova B. The effect of intravenous immunoglobulin on neuromyelitis optica. *Neurologia* 2013;28(2):65-72. DOI: 10.1016/j.nrl.2012.03.014. [Context Link 1] View abstract...
40. Absoud M, et al. A multicentre randomised controlled TRIal of IntraVENous immunoglobulin compared with standard therapy for the treatment of transverse myelitis in adults and children (STRIVE). *Health Technology Assessment* 2017;21(31):1-50. DOI: 10.3310/hta21310. [Context Link 1] View abstract...
41. Banwell B, et al. Diagnosis of myelin oligodendrocyte glycoprotein antibody-associated disease: International MOGAD Panel proposed criteria. *Lancet Neurology* 2023;22(3):268-282. DOI: 10.1016/S1474-4422(22)00431-8. [Context Link 1] View abstract...
42. Cacciaguerra L, Flanagan EP. Updates in NMOSD and MOGAD diagnosis and treatment: a tale of two central nervous system autoimmune inflammatory disorders. *Neurologic Clinics* 2024;42(1):77-114. DOI: 10.1016/j.ncl.2023.06.009. [Context Link 1] View abstract...

43. Wang X, et al. Effectiveness and tolerability of different therapies in preventive treatment of MOG-IgG-associated disorder: A network meta-analysis. *Frontiers in Immunology* 2022;13:Online. DOI: 10.3389/fimmu.2022.953993. [Context Link 1] View abstract...
44. Klein da Costa B, Banwell BL, Sato DK. Treatment of MOG-IgG associated disease in paediatric patients: A systematic review. *Multiple Sclerosis and Related Disorders* 2021;56:Online. DOI: 10.1016/j.msard.2021.103216. [Context Link 1] View abstract...
45. Fadda G, Armangue T, Hacohen Y, Chitnis T, Banwell B. Paediatric multiple sclerosis and antibody-associated demyelination: clinical, imaging, and biological considerations for diagnosis and care. *Lancet Neurology* 2021;20(2):136-149. DOI: 10.1016/S1474-4422(20)30432-4. [Context Link 1] View abstract...
46. Tenenbaum S, Yeh EA, Guthy-Jackson Foundation International Clinical Consortium (GJCF-ICC). Pediatric NMOSD: a review and position statement on approach to work-up and diagnosis. *Frontiers in Pediatrics* 2020;8:Online. DOI: 10.3389/fped.2020.00339. [Context Link 1] View abstract...
47. Hacohen Y, Banwell B. Treatment approaches for MOG-Ab-associated demyelination in children. *Current Treatment Options in Neurology* 2019;21(1):Online. DOI: 10.1007/s11940-019-0541-x. [Context Link 1] View abstract...
48. Bruijstens AL, et al. E.U. paediatric MOG consortium consensus: Part 5 - Treatment of paediatric myelin oligodendrocyte glycoprotein antibody-associated disorders. *European Journal of Paediatric Neurology* 2020;29:41-53. DOI: 10.1016/j.ejpn.2020.10.005. [Context Link 1] View abstract...
49. Lunn MP, Nobile-Orazio E. Immunotherapy for IgM anti-myelin-associated glycoprotein paraprotein-associated peripheral neuropathies. *Cochrane Database of Systematic Reviews* 2016, Issue 10. Art. No.: CD002827. DOI: 10.1002/14651858.CD002827.pub4. [Context Link 1] View abstract...
50. Joint Task Force of the EFNS and the PNS. European Federation of Neurological Societies/Peripheral Nerve Society Guideline on management of paraproteinemic demyelinating neuropathies. Report of a Joint Task Force of the European Federation of Neurological Societies and the Peripheral Nerve Society--first revision. *Journal of the Peripheral Nervous System* 2010;15(3):185-195. DOI: 10.1111/j.1529-8027.2010.00278.x. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
51. Baizabal-Carvallo JF, Jankovic J. Movement disorders in autoimmune diseases. *Movement Disorders* 2012;27(8):935-946. DOI: 10.1002/mds.25011. [Context Link 1, 2] View abstract...
52. Frankovich J, et al. Clinical management of pediatric acute-onset neuropsychiatric syndrome: part II-use of immunomodulatory therapies. *Journal of Child and Adolescent Psychopharmacology* 2017;27(7):574-593. DOI: 10.1089/cap.2016.0148. [Context Link 1]
53. Koopman FS, Beelen A, Gilhus NE, de Visser M, Nollet F. Treatment for postpolio syndrome. *Cochrane Database of Systematic Reviews* 2015, Issue 5. Art. No.: CD007818. DOI: 10.1002/14651858.CD007818.pub3. [Context Link 1] View abstract...
54. Bertolasi L, et al. A randomized controlled trial of IV immunoglobulin in patients with postpolio syndrome. *Journal of the Neurological Sciences* 2013;330(1-2):94-99. DOI: 10.1016/j.jns.2013.04.016. [Context Link 1] View abstract...
55. Friedman DM, et al. Evaluation of fetuses in a study of intravenous immunoglobulin as preventive therapy for congenital heart block: Results of a multicenter, prospective, open-label clinical trial. *Arthritis and Rheumatism* 2010;62(4):1138-1146. DOI: 10.1002/art.27308. [Context Link 1] View abstract...
56. Pisoni CN, et al. Failure of intravenous immunoglobulin to prevent congenital heart block: Findings of a multicenter, prospective, observational study. *Arthritis and Rheumatism* 2010;62(4):1147-1152. DOI: 10.1002/art.27350. [Context Link 1] View abstract...
57. Donofrio MT, et al. Diagnosis and treatment of fetal cardiac disease: a scientific statement from the American Heart Association. *Circulation* 2014;129(21):2183-242. DOI: 10.1161/01.cir.0000437597.44550.5d. [Context Link 1] View abstract...
58. Woon EV, Day A, Bracewell-Milnes T, Male V, Johnson M. Immunotherapy to improve pregnancy outcome in women with abnormal natural killer cell levels/activity and recurrent miscarriage or implantation failure: A systematic review and meta-analysis. *Journal of Reproductive Immunology* 2020;142:103189. DOI: 10.1016/j.jri.2020.103189. [Context Link 1] View abstract...
59. Ata B, Tan SL, Shehata F, Holzer H, Buckett W. A systematic review of intravenous immunoglobulin for treatment of unexplained recurrent miscarriage. *Fertility and Sterility* 2011;95(3):1080-5.e1-e2. DOI: 10.1016/j.fertnstert.2010.12.021. [Context Link 1] View abstract...
60. Christiansen OB, Larsen EC, Egerup P, Lunoe L, Egestad L, Nielsen HS. Intravenous immunoglobulin treatment for secondary recurrent miscarriage: a randomised, double-blind, placebo-controlled trial. *BJOG: An International Journal of Obstetrics and Gynaecology* 2015;122(4):500-508. DOI: 10.1111/1471-0528.13192. [Context Link 1] View abstract...
61. Li J, Chen Y, Liu C, Hu Y, Li L. Intravenous immunoglobulin treatment for repeated IVF/ICSI failure and unexplained infertility: a systematic review and a meta-analysis. *American Journal of Reproductive Immunology : AJRI* 2013;70(6):434-447. DOI: 10.1111/aji.12170. [Context Link 1] View abstract...
62. Cilliers A, Adler AJ, Saloojee H. Anti-inflammatory treatment for carditis in acute rheumatic fever. *Cochrane Database of Systematic Reviews* 2015, (verified by Cochrane 2016 Apr), Issue 5. Art. No.: CD003176. DOI: 10.1002/14651858.CD003176.pub3. [Context Link 1] View abstract...
63. Katz-Agranov N, Khattri S, Zandman-Goddard G. The role of intravenous immunoglobulins in the treatment of rheumatoid arthritis. *Autoimmunity Reviews* 2015;14(8):651-658. DOI: 10.1016/j.autrev.2015.04.003. [Context Link 1] View abstract...
64. Alejandria MM, Lansang MA, Dans LF, Mantaring Iii JB. Intravenous immunoglobulin for treating sepsis, severe sepsis and septic shock. *Cochrane Database of Systematic Reviews* 2013, (verified by Cochrane 2018 Dec), Issue 9. Art. No.: CD001090. DOI: 10.1002/14651858.CD001090.pub2. [Context Link 1, 2] View abstract...
65. Ohlsson A, Lacy JB. Intravenous immunoglobulin for suspected or proven infection in neonates. *Cochrane Database of Systematic Reviews* 2020, Issue 1. Art. No.: CD001239. DOI: 10.1002/14651858.CD001239.pub6. [Context Link 1] View abstract...
66. Franco AC, Torrico AC, Moreira FT, Sa FP, D'Elia HV, Bernardo WM. Adjuvant use of intravenous immunoglobulin in the treatment of neonatal sepsis: a systematic review with a meta-analysis. *Jornal de Pediatria* 2012;88(5):377-383. DOI: doi:10.2223/JPED.2218. [Context Link 1] View abstract...

67. INIS Collaborative Group, et al. Treatment of neonatal sepsis with intravenous immune globulin. *New England Journal of Medicine* 2011;365(13):1201-1211. DOI: 10.1056/NEJMoa1100441. [Context Link 1] View abstract...
68. Ohlsson A, Lacy JB. Intravenous immunoglobulin for preventing infection in preterm and/or low birth weight infants. *Cochrane Database of Systematic Reviews* 2020, Issue 1. Art. No.: CD000361. DOI: 10.1002/14651858.CD000361.pub4. [Context Link 1] View abstract...
69. Takehara K, Ihn H, Sato S. A randomized, double-blind, placebo-controlled trial: intravenous immunoglobulin treatment in patients with diffuse cutaneous systemic sclerosis. *Clinical and Experimental Rheumatology* 2013;31(2 Suppl 76):151-156. [Context Link 1] View abstract...
70. Cantarini L, et al. Intravenous immunoglobulins (IVIg) in systemic sclerosis: a challenging yet promising future. *Immunologic Research* 2015;61(3):326-337. DOI: 10.1007/s12026-014-8615-z. [Context Link 1] View abstract...
71. Foeldvari I, et al. Best clinical practice in the treatment of juvenile systemic sclerosis: expert panel guidance - the result of the International Hamburg Consensus Meeting December 2022. *Expert Review of Clinical Immunology* 2024;20(4):387-404. DOI: 10.1080/1744666X.2023.2298354. [Context Link 1] View abstract...
72. Shehata N, et al. The use of immunoglobulin therapy for patients undergoing solid organ transplantation: an evidence-based practice guideline. *Transfusion Medicine Reviews* 2010;24 Suppl 1:S7-S27. DOI: 10.1016/j.tmr.2009.09.010. [Context Link 1, 2, 3, 4, 5, 6, 7] View abstract...
73. Jordan SC, Toyoda M, Kahwaji J, Vo AA. Clinical aspects of intravenous immunoglobulin use in solid organ transplant recipients. *American Journal of Transplantation* 2011;11(2):196-202. DOI: 10.1111/j.1600-6143.2010.03400.x. [Context Link 1] View abstract...
74. Colvin MM, et al. Antibody-mediated rejection in cardiac transplantation: emerging knowledge in diagnosis and management: a scientific statement from the American Heart Association. *Circulation* 2015;131(18):1608-1639. DOI: 10.1161/CIR.0000000000000093. [Context Link 1] View abstract...
75. Watkins C, Peiris E, Saleh H, Krishnaswamy G. Intravenous immunoglobulin as a potential therapy for refractory urticaria - a review. *Inflammation & Allergy Drug Targets* 2012;11(5):375-381. [Context Link 1] View abstract...
76. Enk A, et al. European Guidelines (S1) on the use of high-dose intravenous immunoglobulin in dermatology. *Journal of the German Society of Dermatology* 2017;15(2):228-241. DOI: 10.1111/ddg.13013. [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10] View abstract...
77. Shimizu T, Morita T, Kumanogoh A. The therapeutic efficacy of intravenous immunoglobulin in anti-neutrophilic cytoplasmic antibody-associated vasculitis: a meta-analysis. *Rheumatology* 2020;59(5):959-967. DOI: 10.1093/rheumatology/kez311. [Context Link 1] View abstract...
78. Fortin PM, Tejani AM, Bassett K, Musini VM. Intravenous immunoglobulin as adjuvant therapy for Wegener's granulomatosis. *Cochrane Database of Systematic Reviews* 2013, Issue 1. Art. No.: CD007057. DOI: 10.1002/14651858.CD007057.pub3. [Context Link 1] View abstract...
79. Arimura Y, et al. Intravenous immunoglobulin for chronic residual peripheral neuropathy in microscopic polyangiitis: A multicentre randomised double-blind trial. *Modern Rheumatology* 2023;33(6):1125-1136. DOI: 10.1093/mr/roac137. [Context Link 1] View abstract...
80. Chung SA, et al. 2021 American College of Rheumatology/Vasculitis Foundation guideline for the management of antineutrophil cytoplasmic antibody-associated vasculitis. *Arthritis & Rheumatology (Hoboken, N.J.)* 2021;73(8):1366-1383. DOI: 10.1002/art.41773. (Reaffirmed 2024 Sep) [Context Link 1, 2] View abstract...
81. Kidney Disease: Improving Global Outcomes (KDIGO) ANCA Vasculitis Work Group. KDIGO 2024 clinical practice guideline for the management of antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis. *Kidney International* 2024;105(3S):S71-S116. DOI: 10.1016/j.kint.2023.10.008. [Context Link 1] View abstract...
82. Iro MA, Martin NG, Absoud M, Pollard AJ. Intravenous immunoglobulin for the treatment of childhood encephalitis. *Cochrane Database of Systematic Reviews* 2017, Issue 10. Art. No.: CD011367. DOI: 10.1002/14651858.CD011367.pub2. [Context Link 1] View abstract...
83. Means RT. Pure red cell aplasia. *Blood* 2016;128(21):2504-2509. DOI: 10.1182/blood-2016-05-717140. [Context Link 1, 2, 3, 4] View abstract...
84. Balasubramanian SK, et al. Rational management approach to pure red cell aplasia. *Haematologica* 2018;103(2):221-230. DOI: 10.3324/haematol.2017.175810. [Context Link 1, 2, 3] View abstract...
85. Qiu J, Soderlund-Venermo M, Young NS. Human parvoviruses. *Clinical Microbiology Reviews* 2017;30(1):43-113. DOI: 10.1128/CMR.00040-16. [Context Link 1] View abstract...
86. Wierda WG, et al. Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma. *NCCN Clinical Practice Guidelines in Oncology [Internet]* National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Mar 26 Accessed at: <https://www.nccn.org/>. [accessed 2024 Apr 29] [Context Link 1, 2, 3]
87. Rotz SJ, et al. International recommendations for screening and preventative practices for long-term survivors of transplantation and cellular therapy: a 2023 update. *Transplantation and Cellular Therapy* 2024;30(4):349-385. DOI: 10.1016/j.jtct.2023.12.001. [Context Link 1, 2] View abstract...
88. Otani IM, et al. Practical guidance for the diagnosis and management of secondary hypogammaglobulinemia: A Work Group Report of the AAAAI Primary Immunodeficiency and Altered Immune Response Committees. *Journal of Allergy and Clinical Immunology* 2022;149(5):1525-1560. DOI: 10.1016/j.jaci.2022.01.025. [Context Link 1, 2, 3] View abstract...
89. Baden LR, et al. Prevention and Treatment of Cancer-Related Infections. *NCCN Clinical Practice Guidelines in Oncology [Internet]* National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Aug Accessed at: <https://www.nccn.org/>. [accessed 2024 Aug 16] [Context Link 1]
90. Loren AW, et al. Hematopoietic Cell Transplantation (HCT). *NCCN Clinical Practice Guidelines in Oncology [Internet]* National Comprehensive Cancer Network (NCCN). v. 1.2024; 2024 Apr 26 Accessed at: <https://www.nccn.org/>. [accessed 2024 Apr 29] [Context Link 1]
91. Jolles S, et al. Treating secondary antibody deficiency in patients with haematological malignancy: European expert consensus. *European Journal of Haematology* 2021;106(4):439-449. DOI: 10.1111/ejh.13580. [Context Link 1, 2] View abstract...
92. Iversen M, et al. Supportive care during pediatric hematopoietic stem cell transplantation: prevention of infections. a report from Workshops on Supportive Care of the Paediatric Diseases Working Party (PDWP) of the European Society for Blood and Marrow Transplantation (EBMT). *Frontiers in Pediatrics* 2021;9:Online. DOI: 10.3389/fped.2021.705179. [Context Link 1] View abstract...

93. Hellmich B, et al. EULAR recommendations for the management of ANCA-associated vasculitis: 2022 update. *Annals of the Rheumatic Diseases* 2023;Online. DOI: 10.1136/ard-2022-223764. [Context Link 1] View abstract...
94. Wijetilleka S, et al. Recommendations for the management of secondary hypogammaglobulinaemia due to B cell targeted therapies in autoimmune rheumatic diseases. *Rheumatology* 2019;58(5):889-896. DOI: 10.1093/rheumatology/key394. [Context Link 1] View abstract...
95. Zelenetz AD, et al. B-Cell Lymphomas. NCCN Clinical Practice Guideline in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Apr 30 Accessed at: <https://www.nccn.org/>. [accessed 2024 May 20] [Context Link 1, 2]
96. Barth M, et al. Pediatric Aggressive Mature B-Cell Lymphomas. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2024; 2024 Apr 08 Accessed at: <https://www.nccn.org/>. [accessed 2024 Apr 29] [Context Link 1]
97. Legault K, et al. McMaster RARE-Bestpractices clinical practice guideline on diagnosis and management of the catastrophic antiphospholipid syndrome. *Journal of Thrombosis and Haemostasis* 2018;Online. DOI: 10.1111/jth.14192. [Context Link 1, 2] View abstract...
98. Garcia D, Erkan D. Diagnosis and management of the antiphospholipid syndrome. *New England Journal of Medicine* 2018;378(21):2010-2021. DOI: 10.1056/NEJMr1705454. [Context Link 1] View abstract...
99. Rodriguez-Pinto I, Espinosa G, Erkan D, Shoenfeld Y, Cervera R, CAPS Registry Project Group. The effect of triple therapy on the mortality of catastrophic anti-phospholipid syndrome patients. *Rheumatology* 2018;57(7):1264-1270. DOI: 10.1093/rheumatology/key082. [Context Link 1] View abstract...
100. Tektonidou MG, et al. EULAR recommendations for the management of antiphospholipid syndrome in adults. *Annals of the Rheumatic Diseases* 2019;78(10):1296-1304. DOI: 10.1136/annrheumdis-2019-215213. [Context Link 1] View abstract...
101. Kianfar N, Dasdar S, Daneshpazhooh M, Aryanian Z, Goodarzi A. A systematic review on efficacy, safety and treatment durability of intravenous immunoglobulin in autoimmune bullous dermatoses: Special focus on indication and combination therapy. *Experimental Dermatology* 2023;32(7):934-944. DOI: 10.1111/exd.14829. [Context Link 1] View abstract...
102. Cakmak SK, Cakmak AI, Gonul M, K I c A, Gul U. Intravenous immunoglobulin therapy in dermatology: an update. *Inflammation & Allergy Drug Targets* 2013;12(2):132-146. [Context Link 1, 2, 3, 4, 5] View abstract...
103. Lytvyn Y, et al. Biologic treatment outcomes in mucous membrane pemphigoid: a systematic review. *Journal of the American Academy of Dermatology* 2022;87(1):110-120. DOI: 10.1016/j.jaad.2020.12.056. [Context Link 1, 2] View abstract...
104. Amagai M, et al. A randomized double-blind trial of intravenous immunoglobulin for bullous pemphigoid. *Journal of Dermatological Science* 2017;85(2):77-84. DOI: 10.1016/j.jdermsci.2016.11.003. [Context Link 1] View abstract...
105. Patel PM, Jones VA, Murray TN, Amber KT. A review comparing international guidelines for the management of bullous pemphigoid, pemphigoid gestationis, Mucous Membrane Pemphigoid, and Epidermolysis Bullosa Acquisita. *American Journal of Clinical Dermatology* 2020;21(4):557-565. DOI: 10.1007/s40257-020-00513-3. [Context Link 1, 2, 3, 4] View abstract...
106. Gurcan HM, Ahmed AR. Current concepts in the treatment of epidermolysis bullosa acquisita. *Expert Opinion on Pharmacotherapy* 2011;12(8):1259-1268. DOI: 10.1517/14656566.2011.549127. [Context Link 1] View abstract...
107. Altman EM. Novel therapies for pemphigus vulgaris. *American Journal of Clinical Dermatology* 2020;21(6):765-782. DOI: 10.1007/s40257-020-00544-w. [Context Link 1] View abstract...
108. Lunemann JD, Quast I, Dalakas MC. Efficacy of intravenous immunoglobulin in neurological diseases. *Neurotherapeutics* 2016;13(1):34-46. DOI: 10.1007/s13311-015-0391-5. [Context Link 1, 2, 3, 4, 5, 6, 7] View abstract...
109. Dalmau J, Graus F. Antibody-mediated encephalitis. *New England Journal of Medicine* 2018;378(9):840-851. DOI: 10.1056/NEJMr1708712. [Context Link 1, 2, 3] View abstract...
110. Abboud H, et al. Autoimmune encephalitis: proposed best practice recommendations for diagnosis and acute management. *Journal of Neurology, Neurosurgery, and Psychiatry* 2021;92(7):757-768. DOI: 10.1136/jnnp-2020-325300. [Context Link 1] View abstract...
111. Bien CG. Management of autoimmune encephalitis. *Current Opinion in Neurology* 2021;34(2):166-171. DOI: 10.1097/WCO.0000000000000909. [Context Link 1, 2] View abstract...
112. Hardy D. Autoimmune encephalitis in children. *Pediatric Neurology* 2022;132:56-66. DOI: 10.1016/j.pediatrneurol.2022.05.004. [Context Link 1, 2, 3, 4] View abstract...
113. Nosadini M, et al. Use and safety of immunotherapeutic management of N-methyl-d-aspartate receptor antibody encephalitis: a meta-analysis. *JAMA Neurology* 2021;78(11):1333-1344. DOI: 10.1001/jamaneurol.2021.3188. [Context Link 1] View abstract...
114. Zuliani L, Graus F, Giometto B, Bien C, Vincent A. Central nervous system neuronal surface antibody associated syndromes: review and guidelines for recognition. *Journal of Neurology, Neurosurgery, and Psychiatry* 2012;83(6):638-645. DOI: 10.1136/jnnp-2011-301237. [Context Link 1] View abstract...
115. Mulhearn B, Bruce IN. Indications for IVIG in rheumatic diseases. *Rheumatology* 2015;54(3):383-391. DOI: 10.1093/rheumatology/keu429. [Context Link 1, 2, 3] View abstract...
116. Hill QA, et al. Guidelines on the management of drug-induced immune and secondary autoimmune, haemolytic anaemia. *British Journal of Haematology* 2017;177(2):208-220. DOI: 10.1111/bjh.14654. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
117. Barcellini W, et al. Clinical heterogeneity and predictors of outcome in primary autoimmune hemolytic anemia: a GIMEMA study of 308 patients. *Blood* 2014;124(19):2930-2936. DOI: 10.1182/blood-2014-06-583021. [Context Link 1] View abstract...
118. Thompson JA, et al. Management of Immunotherapy-Related Toxicities. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2024; 2023 Dec 07 Accessed at: <https://www.nccn.org/>. [accessed 2024 Jan 14] [Context Link 1, 2, 3, 4, 5, 6, 7]
119. Santomaso BD, et al. Management of immune-related adverse events in patients treated with chimeric antigen receptor T-cell therapy: ASCO guideline. *Journal of Clinical Oncology* 2021;39(35):3978-3992. DOI: 10.1200/JCO.21.01992. (Reaffirmed 2024 Jul) [Context Link 1, 2] View abstract...

abstract...

120. Hayden PJ, et al. Management of adults and children receiving CAR T-cell therapy: 2021 best practice recommendations of the European Society for Blood and Marrow Transplantation (EBMT) and the Joint Accreditation Committee of ISCT and EBMT (JACIE) and the European Haematology Association (EHA). *Annals of Oncology* 2022;33(3):259-275. DOI: 10.1016/j.annonc.2021.12.003. [Context Link 1, 2] View abstract...
121. Wat J, Barnett S. Hypogammaglobulinemia after chimeric antigen receptor (CAR) T-cell therapy: characteristics, management, and future directions. *Journal of Allergy and Clinical Immunology. In Practice* 2022;10(2):460-466. DOI: 10.1016/j.jaip.2021.10.037. [Context Link 1] View abstract...
122. Maus MV, et al. Society for Immunotherapy of Cancer (SITC) clinical practice guideline on immune effector cell-related adverse events. *Journal for Immunotherapy of Cancer* 2020;8(2):Online. DOI: 10.1136/jitc-2020-001511. [Context Link 1] View abstract...
123. Jain T, et al. Use of chimeric antigen receptor T cell therapy in clinical practice for relapsed/refractory aggressive B cell non-Hodgkin lymphoma: an expert panel opinion from the American Society for Transplantation and Cellular Therapy. *Biology of Blood and Marrow Transplantation* 2019;25(12):2305-2321. DOI: 10.1016/j.bbmt.2019.08.015. [Context Link 1] View abstract...
124. Anderson LD, et al. Chimeric antigen receptor T cell therapy for myeloma: where are we now and what is needed to move chimeric antigen receptor T cells forward to earlier lines of therapy? Expert panel opinion from the American Society for Transplantation and Cellular Therapy. *Transplantation and Cellular Therapy* 2024;30(1):17-37. DOI: 10.1016/j.jtct.2023.10.022. [Context Link 1] View abstract...
125. Gammaked [immune globulin injection (human), 10% caprylate/chromatography purified]. Physician Prescribing Information [Internet] Grifols Therapeutics, Inc. 2020 Jan Accessed at: <https://www.gammaked.com/>. [created 2003; accessed 2024 Sep 09] [Context Link 1, 2, 3, 4, 5, 6]
126. Gamunex-C [immune globulin injection (human), 10% caprylate/chromatography purified]. Physician Prescribing Information [Internet] Grifols Therapeutics, Inc. 2022 Apr Accessed at: <https://www.gamunex-c.com/en/hcp/cidp>. [created 2003; accessed 2024 Sep 25] [Context Link 1, 2, 3, 4, 5]
127. Privigen immune globulin intravenous (human), 10% liquid. Physician Prescribing Information [Internet] CSL Behring LLC. 2022 Mar Accessed at: <https://www.privigen.com/>. [created 2007; accessed 2024 Oct 07] [Context Link 1, 2, 3, 4]
128. Van den Bergh PYK, et al. European Academy of Neurology/Peripheral Nerve Society guideline on diagnosis and treatment of chronic inflammatory demyelinating polyradiculoneuropathy: Report of a joint Task Force-Second revision. *European Journal of Neurology* 2021;28(11):3556-3583. DOI: 10.1111/ene.14959. [Context Link 1, 2] View abstract...
129. Gammagard Liquid, immune globulin infusion (human). Physician Prescribing Information [Internet] Baxalta US Inc. 2024 Jan Accessed at: <https://www.gammagard.com/primary-immunodeficiency/patient/>. [created 2005; accessed 2024 Sep 25] [Context Link 1, 2, 3, 4, 5, 6, 7, 8]
130. Hizentra®, immune globulin subcutaneous (human), 20% Liquid. Physician Prescribing Information [Internet] CSL Behring LLC Pharmacovigilance. 2023 Apr Accessed at: <https://www.hizentra.com/>. [accessed 2024 Sep 25] [Context Link 1, 2]
131. van Schaik IN, et al. Subcutaneous immunoglobulin for maintenance treatment in chronic inflammatory demyelinating polyneuropathy (PATH): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Neurology* 2018;17(1):35-46. DOI: 10.1016/S1474-4422(17)30378-2. [Context Link 1, 2] View abstract...
132. HyQvia [immune globulin infusion 10% (human) with recombinant human hyaluronidase]. Physician Prescribing Information [Internet] Baxalta Inc. 2024 Jan Accessed at: <https://www.hyqvia.com/>. [created 2014; accessed 2024 Sep 25] [Context Link 1, 2, 3, 4]
133. Bus SR, de Haan RJ, Vermeulen M, van Schaik IN, Eftimov F. Intravenous immunoglobulin for chronic inflammatory demyelinating polyradiculoneuropathy. *Cochrane Database of Systematic Reviews* 2024, Issue 2. Art. No.: CD001797. DOI: 10.1002/14651858.CD001797.pub4. [Context Link 1, 2] View abstract...
134. Kiehmet J, Staudt M, Ulm L, Unterwalder N, Meisel A, Meisel C. Circulating lymphocyte and T memory subsets in glucocorticosteroid versus IVIG treated patients with CIDP. *Journal of Neuroimmunology* 2015;283:17-22. DOI: 10.1016/j.jneuroim.2015.03.023. [Context Link 1] View abstract...
135. McMillan HJ, Kang PB, Jones HR, Darras BT. Childhood chronic inflammatory demyelinating polyradiculoneuropathy: combined analysis of a large cohort and eleven published series. *Neuromuscular Disorders: NMD* 2013;23(2):103-111. DOI: 10.1016/j.nmd.2012.09.008. [Context Link 1] View abstract...
136. Bunschoten C, Jacobs BC, Van den Bergh PYK, Cornblath DR, van Doorn PA. Progress in diagnosis and treatment of chronic inflammatory demyelinating polyradiculoneuropathy. *Lancet Neurology* 2019;18(8):784-794. DOI: 10.1016/S1474-4422(19)30144-9. [Context Link 1] View abstract...
137. Fehmi J, Bellanti R, Misbah SA, Bhattacharjee A, Rinaldi S. Treatment of CIDP. *Practical Neurology* 2023;23(1):46-53. DOI: 10.1136/pn-2021-002991. [Context Link 1] View abstract...
138. Kuwabara S, et al. Intravenous immunoglobulin for maintenance treatment of chronic inflammatory demyelinating polyneuropathy: a multicentre, open-label, 52-week phase III trial. *Journal of Neurology, Neurosurgery, and Psychiatry* 2017;88(10):832-838. DOI: 10.1136/jnnp-2017-316427. [Context Link 1] View abstract...
139. Octagam 10% [immune globulin intravenous (human)] liquid solution for intravenous administration. Physician Prescribing Information [Internet] Octapharma. 2022 Mar Accessed at: <https://octagamusa.com/octagam10/>. [created 2014; accessed 2024 Sep 30] [Context Link 1, 2, 3]
140. Gordon PA, Winer JB, Hoogendijk JE, Choy EH. Immunosuppressant and immunomodulatory treatment for dermatomyositis and polymyositis. *Cochrane Database of Systematic Reviews* 2012, (verified by Cochrane 2016 Aug), Issue 9. Art. No.: CD003643. DOI: 10.1002/14651858.CD003643.pub4. [Context Link 1, 2] View abstract...
141. Gandiga PC, Ghetie D, Anderson E, Aggrawal R. Intravenous immunoglobulin in idiopathic inflammatory myopathies: a practical guide for clinical use. *Current Rheumatology Reports* 2023;25(8):152-168. DOI: 10.1007/s11926-023-01105-w. [Context Link 1, 2] View abstract...
142. Dalakas MC. Inflammatory muscle diseases. *New England Journal of Medicine* 2015;372(18):1734-1747. DOI: 10.1056/NEJMr1402225. [Context Link 1] View abstract...

143. Dalakas MC. Autoimmune inflammatory myopathies. *Handbook of Clinical Neurology* 2023;195:425-460. DOI: 10.1016/B978-0-323-98818-6.00023-6. [Context Link 1] View abstract...
144. Aggarwal R, et al. Trial of intravenous immune globulin in dermatomyositis. *New England Journal of Medicine* 2022;387(14):1264-1278. DOI: 10.1056/NEJMoa2117912. [Context Link 1] View abstract...
145. Anh-Tu Hoa S, Hudson M. Critical review of the role of intravenous immunoglobulins in idiopathic inflammatory myopathies. *Seminars in Arthritis and Rheumatism* 2017;46(4):488-508. DOI: 10.1016/j.semarthrit.2016.07.014. [Context Link 1] View abstract...
146. Van Der Lugt NM, et al. Neonatal outcome in alloimmune thrombocytopenia after maternal treatment with intravenous immunoglobulin. *Blood Transfusion* 2015;13(1):66-71. DOI: 10.2450/2014.0309-13. [Context Link 1, 2, 3] View abstract...
147. Winkelhorst D, et al. Antenatal management in fetal and neonatal alloimmune thrombocytopenia: a systematic review. *Blood* 2017;129(11):1538-1547. DOI: 10.1182/blood-2016-10-739656. [Context Link 1, 2] View abstract...
148. Batton E, Leibel SL. Immune-mediated neonatal thrombocytopenia. *NeoReviews* 2022;23(7):e462-e471. DOI: 10.1542/neo.23-7-e462. [Context Link 1] View abstract...
149. Rayment R, Brunskill SJ, Soothill PW, Roberts DJ, Bussell JB, Murphy MF. Antenatal interventions for fetomaternal alloimmune thrombocytopenia. *Cochrane Database of Systematic Reviews* 2011, Issue 5. Art. No.: CD004226. DOI: 10.1002/14651858.CD004226.pub3. [Context Link 1] View abstract...
150. Willison HJ, Jacobs BC, van Doorn PA. Guillain-Barre syndrome. *Lancet* 2016;388(10045):717-727. DOI: 10.1016/S0140-6736(16)00339-1. [Context Link 1] View abstract...
151. Morales-Ruiz V, Juarez-Vaquera VH, Rosetti-Sciutto M, Sanchez-Munoz F, Adalid-Peralta L. Efficacy of intravenous immunoglobulin in autoimmune neurological diseases. Literature systematic review and meta-analysis. *Autoimmunity Reviews* 2022;21(3):Online. DOI: 10.1016/j.autrev.2021.103019. [Context Link 1, 2] View abstract...
152. van Doorn PA, et al. European Academy of Neurology/Peripheral Nerve Society Guideline on diagnosis and treatment of Guillain-Barre syndrome. *European Journal of Neurology* 2023;30(12):3646-3674. DOI: 10.1111/ene.16073. [Context Link 1, 2] View abstract...
153. Ortiz-Salas P, Velez-Van-Meerbeke A, Galvis-Gomez CA, Rodriguez Q JH. Human immunoglobulin versus plasmapheresis in Guillain-Barre syndrome and myasthenia gravis: a meta-analysis. *Journal of Clinical Neuromuscular Disease* 2016;18(1):1-11. DOI: 10.1097/CND.000000000000119. [Context Link 1] View abstract...
154. Hughes RA, Swan AV, van Doorn PA. Intravenous immunoglobulin for Guillain-Barre syndrome. *Cochrane Database of Systematic Reviews* 2014, (verified by Cochrane 2019 Sep), Issue 9. Art. No.: CD002063. DOI: 10.1002/14651858.CD002063.pub6. [Context Link 1] View abstract...
155. Walgaard C, et al. Second intravenous immunoglobulin dose in patients with Guillain-Barre syndrome with poor prognosis (SID-GBS): a double-blind, randomised, placebo-controlled trial. *Lancet Neurology* 2021;20(4):275-283. DOI: 10.1016/S1474-4422(20)30494-4. [Context Link 1] View abstract...
156. Korinthenberg R, et al. Diagnosis and treatment of Guillain-Barre Syndrome in childhood and adolescence: An evidence- and consensus-based guideline. *European Journal of Paediatric Neurology* 2020;25:5-16. DOI: 10.1016/j.ejpn.2020.01.003. [Context Link 1] View abstract...
157. Liu J, Wang LN, McNicol ED. Pharmacological treatment for pain in Guillain-Barre syndrome. *Cochrane Database of Systematic Reviews* 2015, (verified by Cochrane 2019 Feb), Issue 4. Art. No.: CD009950. DOI: 10.1002/14651858.CD009950.pub3. [Context Link 1] View abstract...
158. Kumar SK, et al. Multiple Myeloma. *NCCN Clinical Practice Guidelines in Oncology* [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 Apr 26 Accessed at: <https://www.nccn.org/>. [accessed 2024 Apr 29] [Context Link 1, 2]
159. Gammagard S/D, immune globulin intravenous (human). Physician Prescribing Information [Internet] Baxter Healthcare Corporation. 2023 Mar Accessed at: <https://www.takeda.com/en-us/what-we-do/product-portfolio#section-g>. [created 1994; accessed 2024 Sep 25] [Context Link 1, 2, 3, 4, 5]
160. Inaba H, et al. Pediatric Acute Lymphoblastic Leukemia. *NCCN Clinical Practice Guidelines in Oncology* [Internet] National Comprehensive Cancer Network (NCCN). v. 6.2024; 2024 Aug Accessed at: <https://www.nccn.org/>. [accessed 2024 Aug 16] [Context Link 1]
161. Shah B, et al. Acute Lymphoblastic Leukemia. *NCCN Clinical Practice Guidelines in Oncology* [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Jul Accessed at: <https://www.nccn.org/>. [accessed 2024 Sep 19] [Context Link 1]
162. Chai KL, et al. Interventions to reduce infections in patients with hematological malignancies: a systematic review and meta-analysis. *Blood Advances* 2023;7(1):20-31. DOI: 10.1182/bloodadvances.2022008073. [Context Link 1] View abstract...
163. Diamond Y, et al. Infections in children following chimeric antigen receptor T-cell therapy for B-cell acute lymphoblastic leukemia. *Transplant Infectious Disease* 2023;25(6):Online. DOI: 10.1111/tid.14202. [Context Link 1] View abstract...
164. Rodriguez-Otero P, et al. International Myeloma Working Group immunotherapy committee consensus guidelines and recommendations for optimal use of T-cell-engaging bispecific antibodies in multiple myeloma. *Lancet Oncology* 2024;25(5):e205-e216. DOI: 10.1016/S1470-2045(24)00043-3. [Context Link 1] View abstract...
165. Ludwig H, et al. Prevention and management of adverse events during treatment with bispecific antibodies and CAR T cells in multiple myeloma: a consensus report of the European Myeloma Network. *Lancet Oncology* 2023;24(6):e255-e269. DOI: 10.1016/S1470-2045(23)00159-6. [Context Link 1, 2] View abstract...
166. Jolles S, et al. Secondary antibody deficiency in chronic lymphocytic leukemia and non-Hodgkin lymphoma: recommendations from an international expert panel. *Blood Reviews* 2023;58:Online. DOI: 10.1016/j.blre.2022.101020. [Context Link 1]
167. Kemper AR, et al. Clinical practice guideline revision: management of hyperbilirubinemia in the newborn infant 35 or more weeks of gestation. *Pediatrics* 2022:e2022058859. DOI: 10.1542/peds.2022-058859. (Reaffirmed 2024 Aug) [Context Link 1, 2, 3] View abstract...
168. Louis D, More K, Oberoi S, Shah PS. Intravenous immunoglobulin in isoimmune haemolytic disease of newborn: an updated systematic review and meta-analysis. *Archives of Disease in Childhood--Fetal and Neonatal Edition* 2014;99(4):F325-F331. DOI: 10.1136/archdischild-2013-304878. [Context Link 1, 2] View abstract...

169. van Klink JM, et al. Immunoglobulins in neonates with Rhesus hemolytic disease of the fetus and newborn: long-term outcome in a randomized trial. *Fetal Diagnosis and Therapy* 2016;39(3):209-213. DOI: 10.1159/000434718. [Context Link 1, 2] View abstract...
170. Zwieters C, Scheffer-Rath ME, Lopriore E, de Haas M, Liley HG. Immunoglobulin for alloimmune hemolytic disease in neonates. *Cochrane Database of Systematic Reviews* 2018, Issue 3. Art. No.: CD003313. DOI: 10.1002/14651858.CD003313.pub2. [Context Link 1] View abstract...
171. Panch SR, Montemayor-Garcia C, Klein HG. Hemolytic transfusion reactions. *New England Journal of Medicine* 2019;381(2):150-162. DOI: 10.1056/NEJMra1802338. [Context Link 1, 2] View abstract...
172. Delaney M, et al. Transfusion reactions: prevention, diagnosis, and treatment. *Lancet* 2016;388(10061):2825-2836. DOI: 10.1016/S0140-6736(15)01313-6. [Context Link 1, 2, 3, 4] View abstract...
173. Chou ST, et al. American Society of Hematology 2020 guidelines for sickle cell disease: transfusion support. *Blood Advances* 2020;4(2):327-355. DOI: 10.1182/bloodadvances.2019001143. [Context Link 1, 2] View abstract...
174. Atteritano M, et al. Haemophagocytic syndrome in rheumatic patients. A systematic review. *European Review for Medical and Pharmacological Sciences* 2012;16(10):1414-1424. [Context Link 1] View abstract...
175. La Rosee P, et al. Recommendations for the management of hemophagocytic lymphohistiocytosis in adults. *Blood* 2019;133(23):2465-2477. DOI: 10.1182/blood.2018894618. [Context Link 1, 2] View abstract...
176. Lehmborg K, et al. Consensus recommendations for the diagnosis and management of hemophagocytic lymphohistiocytosis associated with malignancies. *Haematologica* 2015;100(8):997-1004. DOI: 10.3324/haematol.2015.123562. [Context Link 1] View abstract...
177. Shakoory B, et al. The 2022 EULAR/ACR points to consider at the early stages of diagnosis and management of suspected haemophagocytic lymphohistiocytosis/macrophage activation syndrome (HLH/MAS). *Annals of the Rheumatic Diseases* 2023;Online. DOI: 10.1136/ard-2023-224123. [Context Link 1, 2] View abstract...
178. Tan CJ, Ng ZQ, Bhattacharyya R, Sultana R, Lee JH. Treatment and mortality of hemophagocytic lymphohistiocytosis in critically ill children: A systematic review and meta-analysis. *Pediatric Blood and Cancer* 2023;70(3):Online. DOI: 10.1002/pbc.30122. [Context Link 1] View abstract...
179. Pastori D, Esposito A, Mezzaroma I. Immunomodulatory effects of intravenous immunoglobulins (IVIGs) in HIV-1 disease: a systematic review. *International Reviews of Immunology* 2011;30(1):44-66. DOI: 10.3109/08830185.2010.529975. [Context Link 1, 2, 3, 4] View abstract...
180. Mofenson LM, et al. Guidelines for the Prevention and Treatment of Opportunistic Infections among HIV-exposed and HIV-infected children: recommendations from CDC, the National Institutes of Health, the HIV Medicine Association of the Infectious Diseases Society of America, the Pediatric Infectious Diseases Society, and the American Academy of Pediatrics. *MMWR - Recommendations and Reports* 2009;58(RR-11):1-166. [Context Link 1, 2] View abstract...
181. Al-Samkari H, Kuter DJ. Immune thrombocytopenia in adults: modern approaches to diagnosis and treatment. *Seminars in Thrombosis and Hemostasis* 2020;46(3):275-288. DOI: 10.1055/s-0039-1700512. [Context Link 1] View abstract...
182. Neuner C, et al. American Society of Hematology 2019 guidelines for immune thrombocytopenia. *Blood Advances* 2019;3(23):3829-3866. DOI: 10.1182/bloodadvances.2019000966. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
183. Flebogamma® 10% DIF immune globulin intravenous (human). Physician Prescribing Information [Internet] Grifols Therapeutics, Inc. 2019 Sep Accessed at: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=f9bf6322-75b4-4ccf-9a4b-4724203f5eef>. [created 2010; accessed 2024 Sep 25] [Context Link 1, 2, 3]
184. Gammaplex 5% immune globulin intravenous (human), 5% liquid, for intravenous use. Physician Prescribing Information [Internet] Bio Products Laboratory, Inc. 2019 Sep Accessed at: <https://www.gammaplex.com/>. [created 2009; accessed 2024 Sep 25] [Context Link 1, 2]
185. Gammaplex 10% Immune Globulin Intravenous [Human]. Physician Prescribing Information [Internet] BPL, Inc. 2021 Nov Accessed at: <https://www.gammaplex.com/>. [created 2017; accessed 2024 Sep 25] [Context Link 1]
186. Qin YH, Zhou TB, Su LN, Lei FY, Zhao YJ, Huang WF. The efficacy of different dose intravenous immunoglobulin in treating acute idiopathic thrombocytopenic purpura: a meta-analysis of 13 randomized controlled trials. *Blood Coagulation & Fibrinolysis* 2010;21(8):713-721. DOI: 10.1097/MBC.0b013e3283401490. [Context Link 1] View abstract...
187. Lioger B, Maillot F, Ternant D, Passot C, Paintaud G, Bejan-Angoulvant T. Efficacy and safety of anti-D immunoglobulins versus intravenous immunoglobulins for immune thrombocytopenia in children: systematic review and meta-analysis of randomized controlled trials. *Journal of Pediatrics* 2019;204:225-233.e8. DOI: 10.1016/j.jpeds.2018.07.065. [Context Link 1] View abstract...
188. Heitink-Polle KMJ, et al. Intravenous immunoglobulin vs observation in childhood immune thrombocytopenia: a randomized controlled trial. *Blood* 2018;132(9):883-891. DOI: 10.1182/blood-2018-02-830844. [Context Link 1] View abstract...
189. Russo G, et al. Recommendations for the management of acute immune thrombocytopenia in children. A Consensus Conference from the Italian Association of Pediatric Hematology and Oncology. *Blood Transfusion* 2024;22(3):253-265. DOI: 10.2450/BloodTransfus.501. [Context Link 1] View abstract...
190. Schneider BJ, et al. Management of immune-related adverse events in patients treated with immune checkpoint inhibitor therapy: ASCO guideline update. *Journal of Clinical Oncology* 2021;39(36):4073-4126. DOI: 10.1200/JCO.21.01440. [Context Link 1, 2, 3] View abstract...
191. Kostine M, et al. EULAR points to consider for the diagnosis and management of rheumatic immune-related adverse events due to cancer immunotherapy with checkpoint inhibitors. *Annals of the Rheumatic Diseases* 2021;80(1):36-48. DOI: 10.1136/annrheumdis-2020-217139. [Context Link 1, 2] View abstract...
192. Haanen J, et al. Management of toxicities from immunotherapy: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. *Annals of Oncology* 2022;Online. DOI: 10.1016/j.annonc.2022.10.001. [Context Link 1, 2, 3] View abstract...
193. Choi J, et al. Multinational Association of Supportive Care in Cancer (MASCC) 2020 clinical practice recommendations for the management of severe dermatological toxicities from checkpoint inhibitors. *Supportive Care in Cancer* 2020;28(12):6119-6128. DOI: 10.1007/s00520-020-05706-4. [Context Link 1] View abstract...

194. Suarez-Almazor ME, et al. Multinational Association of Supportive Care in Cancer (MASCC) 2020 clinical practice recommendations for the management of immune-mediated cardiovascular, rheumatic, and renal toxicities from checkpoint inhibitors. *Supportive Care in Cancer* 2020;28(12):6159-6173. DOI: 10.1007/s00520-020-05710-8. (Reaffirmed 2024 Jul) [Context Link 1] View abstract...
195. Jayan A, Mammen AL, Suarez-Almazor ME. Immune checkpoint inhibitor-induced myositis. *Rheumatic Diseases Clinics of North America* 2024;50(2):281-290. DOI: 10.1016/j.rdc.2024.02.003. [Context Link 1] View abstract...
196. Narayanaswami P, et al. International consensus guidance for management of myasthenia gravis: 2020 update. *Neurology* 2021;96(3):114-122. DOI: 10.1212/WNL.00000000000011124. [Context Link 1] View abstract...
197. Brahmer JR, et al. Society for Immunotherapy of Cancer (SITC) clinical practice guideline on immune checkpoint inhibitor-related adverse events. *Journal for Immunotherapy of Cancer* 2021;9(6):Online. DOI: 10.1136/jitc-2021-002435. [Context Link 1] View abstract...
198. Nelson NP, et al. Update: Recommendations of the advisory committee on immunization practices for use of hepatitis A vaccine for postexposure prophylaxis and for preexposure prophylaxis for international travel. *MMWR--Morbidity and Mortality Weekly Report* 2018;67(43):1216-1220. DOI: 10.15585/mmwr.mm6743a5. [Context Link 1, 2, 3, 4] View abstract...
199. McLean HQ, Fiebelkorn AP, Temte JL, Wallace GS, Centers for Disease Control and Prevention. Prevention of measles, rubella, congenital rubella syndrome, and mumps, 2013: summary recommendations of the Advisory Committee on Immunization Practices (ACIP). *MMWR - Recommendations and Reports* 2013;62(RR-04):1-34. [Context Link 1, 2, 3, 4] View abstract...
200. Young MK. The indications and safety of polyvalent immunoglobulin for post-exposure prophylaxis of hepatitis A, rubella and measles. *Human Vaccines and Immunotherapeutics* 2019;15(9):2060-2065. DOI: 10.1080/21645515.2019.1621148. [Context Link 1, 2, 3, 4] View abstract...
201. Young MK, Cripps AW, Nimmo GR, van Driel ML. Post-exposure passive immunisation for preventing rubella and congenital rubella syndrome. *Cochrane Database of Systematic Reviews* 2015, Art. No.: CD010586. DOI: 10.1002/14651858.CD010586.pub2. [Context Link 1] View abstract...
202. Marin M, et al. Prevention of varicella: recommendations of the Advisory Committee on Immunization Practices (ACIP). *MMWR - Recommendations and Reports* 2007;56(RR-4):1-40. [Context Link 1, 2] View abstract...
203. Bader MS, McKinsey DS. Postexposure prophylaxis for common infectious diseases. *American Family Physician* 2013;88(1):25-32. [Context Link 1, 2, 3] View abstract...
204. Nelson NP. Updated dosing instructions for immune globulin (human) gamaSTAN S/D for hepatitis A virus prophylaxis. *MMWR--Morbidity and Mortality Weekly Report* 2017;66(36):959-960. DOI: 10.15585/mmwr.mm6636a5. [Context Link 1] View abstract...
205. Flebogamma 5% DIF (immune globulin intravenous [human]). Physician Prescribing Information [Internet] Grifols Therapeutics, Inc. 2024 Aug Accessed at: <https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=2cf22c72-64ac-45be-b7c6-d20340730096>. [created 2006; accessed 2024 Sep 05] [Context Link 1, 2, 3]
206. Centers for Disease Control and Prevention (CDC). Updated recommendations for use of VarizIG--United States, 2013. *MMWR--Morbidity and Mortality Weekly Report* 2013;62(28):574-6. [Context Link 1] View abstract...
207. Singh S, Sharma A, Rahman MM, Kasniya G, Maheshwari A, Boppana SB. Congenital and perinatal varicella infections. *Newborn : Official Journal of the Global Newborn Society* 2022;1(3):278-286. DOI: 10.5005/jp-journals-11002-0040. [Context Link 1] View abstract...
208. Shah SS, Marin M, LaRussa PS, Gershon AA. Varicella-zoster virus. In: Kliegman RM, et al., editors. *Nelson Textbook of Pediatrics*. 22nd ed. Elsevier; 2025:1999-2005. [Context Link 1]
209. McCrindle BW, et al. Diagnosis, treatment, and long-term management of Kawasaki disease: a scientific statement for health professionals from the American Heart Association. *Circulation* 2017;135(17):e927-e999. DOI: 10.1161/CIR.0000000000000484. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
210. Fraison JB, et al. Kawasaki disease in adults: Observations in France and literature review. *Autoimmunity Reviews* 2016;15(3):242-249. DOI: 10.1016/j.autrev.2015.11.010. [Context Link 1] View abstract...
211. Newburger JW, Takahashi M, Burns JC. Kawasaki Disease. *Journal of the American College of Cardiology* 2016;67(14):1738-1749. DOI: 10.1016/j.jacc.2015.12.073. [Context Link 1, 2] View abstract...
212. Shulman ST. Intravenous immunoglobulin for the treatment of Kawasaki disease. *Pediatric Annals* 2017;46(1):e25-e28. DOI: 10.3928/19382359-20161212-01. [Context Link 1, 2] View abstract...
213. Friedman KG, Jone PN. Update on the management of Kawasaki disease. *Pediatric Clinics of North America* 2020;67(5):811-819. DOI: 10.1016/j.pcl.2020.06.002. [Context Link 1] View abstract...
214. Dimitriades VR, Brown AG, Gedalia A. Kawasaki disease: pathophysiology, clinical manifestations, and management. *Current Rheumatology Reports* 2014;16(6):423. DOI: 10.1007/s11926-014-0423-x. [Context Link 1] View abstract...
215. Eleftheriou D, Levin M, Shingadia D, Tulloh R, Klein NJ, Brogan PA. Management of Kawasaki disease. *Archives of Disease in Childhood* 2014;99(1):74-83. DOI: 10.1136/archdischild-2012-302841. [Context Link 1] View abstract...
216. Marchesi A, et al. Revised recommendations of the Italian Society of Pediatrics about the general management of Kawasaki disease. *Italian Journal of Pediatrics* 2021;47(1):16. DOI: 10.1186/s13052-021-00962-4. [Context Link 1] View abstract...
217. Broderick C, Kobayashi S, Suto M, Ito S, Kobayashi T. Intravenous immunoglobulin for the treatment of Kawasaki disease. *Cochrane Database of Systematic Reviews* 2023, Issue 1. Art. No.: CD014884. DOI: 10.1002/14651858.CD014884.pub2. [Context Link 1] View abstract...
218. Burns JC, et al. Infliximab versus second intravenous immunoglobulin for treatment of resistant Kawasaki disease in the USA (KIDCARE): a randomised, multicentre comparative effectiveness trial. *Lancet Child and Adolescent Health* 2021;5(12):852-861. DOI: 10.1016/S2352-4642(21)00270-4. [Context Link 1] View abstract...
219. Montgomery RA, et al. Desensitization in HLA-incompatible kidney recipients and survival. *New England Journal of Medicine* 2011;365(4):318-326. DOI: 10.1056/NEJMoa1012376. [Context Link 1, 2, 3] View abstract...

220. Sadaka B, Alloway RR, Woodle ES. Management of antibody-mediated rejection in transplantation. *Surgical Clinics of North America* 2013;93(6):1451-1466. DOI: 10.1016/j.suc.2013.08.002. [Context Link 1] View abstract...
221. Cooper JE, Gralla J, Klem P, Chan L, Wiseman AC. High dose intravenous immunoglobulin therapy for donor-specific antibodies in kidney transplant recipients with acute and chronic graft dysfunction. *Transplantation* 2014;97(12):1253-1259. DOI: 10.1097/01.TP.0000443226.74584.03. [Context Link 1] View abstract...
222. Marfo K, Lu A, Ling M, Akalin E. Desensitization protocols and their outcome. *Clinical Journal of the American Society of Nephrology* 2011;6(4):922-936. DOI: 10.2215/CJN.08140910. [Context Link 1] View abstract...
223. Rosenfeld MR, Dalmau J. Diagnosis and management of paraneoplastic neurologic disorders. *Current Treatment Options in Oncology* 2013;14(4):528-538. DOI: 10.1007/s11864-013-0249-1. [Context Link 1] View abstract...
224. Maddison P. Treatment in Lambert-Eaton myasthenic syndrome. *Annals of the New York Academy of Sciences* 2012;1275:78-84. DOI: 10.1111/j.1749-6632.2012.06769.x. [Context Link 1] View abstract...
225. Wiendl H, et al. Guideline for the management of myasthenic syndromes. *Therapeutic Advances in Neurological Disorders* 2023;16:Online. DOI: 10.1177/17562864231213240. [Context Link 1, 2] View abstract...
226. Keogh M, Sedehizadeh S, Maddison P. Treatment for Lambert-Eaton myasthenic syndrome. *Cochrane Database of Systematic Reviews* 2011, Issue 2. Art. No.: CD003279. DOI: 10.1002/14651858.CD003279.pub3. [Context Link 1] View abstract...
227. Giometto B, Vitaliani R, Lindeck-Pozza E, Grisold W, Vedeler C. Treatment for paraneoplastic neuropathies. *Cochrane Database of Systematic Reviews* 2012, Issue 12. Art. No.: CD007625. DOI: 10.1002/14651858.CD007625.pub2. [Context Link 1] View abstract...
228. Joint Task Force of the EFNS and the PNS. European Federation of Neurological Societies/Peripheral Nerve Society guideline on management of multifocal motor neuropathy. Report of a joint task force of the European Federation of Neurological Societies and the Peripheral Nerve Society--first revision. *Journal of the Peripheral Nervous System* 2010;15(4):295-301. DOI: 10.1111/j.1529-8027.2010.00290.x. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
229. Briani C, Cocito D, Campagnolo M, Doneddu PE, Nobile-Orazio E. Update on therapy of chronic immune-mediated neuropathies. *Neurological Sciences* 2022;43(Suppl 2):S605-S614. DOI: 10.1007/s10072-020-04998-y. [Context Link 1] View abstract...
230. Keddle S, Eftimov F, van den Berg LH, Brassington R, de Haan RJ, van Schaik IN. Immunoglobulin for multifocal motor neuropathy. *Cochrane Database of Systematic Reviews* 2022, Issue 1. Art. No.: CD004429. DOI: 10.1002/14651858.CD004429.pub3. [Context Link 1, 2] View abstract...
231. Yeh EA, Weinstock-Guttman B. The management of pediatric multiple sclerosis. *Journal of Child Neurology* 2012;27(11):1384-1393. DOI: 10.1177/0883073812452785. [Context Link 1] View abstract...
232. Russell MD, et al. British Society for Rheumatology guideline on prescribing drugs in pregnancy and breastfeeding: immunomodulatory anti-rheumatic drugs and corticosteroids. *Rheumatology* 2023;62(4):e48-e88. DOI: 10.1093/rheumatology/keac551. (Reaffirmed 2024 Sep 12) [Context Link 1] View abstract...
233. Bhimraj A, et al. Infectious Diseases Society of America Guidelines on the Treatment and Management of Patients with COVID-19. v. 11.0.0 Part 1 [Internet] Infectious Diseases Society of America. 2024 Aug 12 Accessed at: <https://www.idsociety.org/>. [created 2023 Jun 26; accessed 2024 Sep 26] [Context Link 1, 2, 3]
234. Henderson LA, et al. American College of Rheumatology clinical guidance for multisystem inflammatory syndrome in children associated with SARS-CoV-2 and hyperinflammation in pediatric COVID-19: version 3. *Arthritis & Rheumatology (Hoboken, N.J.)* 2022;74(4):e1-e20. DOI: 10.1002/art.42062. [Context Link 1, 2, 3, 4] View abstract...
235. Nijman RG, et al. Pediatric inflammatory multisystem syndrome: statement by the pediatric section of the European Society for Emergency Medicine and European Academy of Pediatrics. *Frontiers in Pediatrics* 2020;8:490. DOI: 10.3389/fped.2020.00490. [Context Link 1, 2] View abstract...
236. Harwood R, et al. A national consensus management pathway for paediatric inflammatory multisystem syndrome temporally associated with COVID-19 (PIMS-TS): results of a national Delphi process. *Lancet Child and Adolescent Health* 2021;5(2):133-141. DOI: 10.1016/S2352-4642(20)30304-7. [Context Link 1, 2] View abstract...
237. Zachariah P. COVID-19 in children. *Infectious Disease Clinics of North America* 2022;36(1):1-14. DOI: 10.1016/j.idc.2021.11.002. [Context Link 1] View abstract...
238. RECOVERY Collaborative Group. Immunomodulatory therapy in children with paediatric inflammatory multisystem syndrome temporally associated with SARS-CoV-2 (PIMS-TS, MIS-C; RECOVERY): a randomised, controlled, open-label, platform trial. *Lancet Child and Adolescent Health* 2024;8(3):190-200. DOI: 10.1016/S2352-4642(23)00316-4. [Context Link 1] View abstract...
239. Welzel T, et al. Methylprednisolone versus intravenous immunoglobulins in children with paediatric inflammatory multisystem syndrome temporally associated with SARS-CoV-2 (PIMS-TS): an open-label, multicentre, randomised trial. *Lancet Child and Adolescent Health* 2023;7(4):238-248. DOI: 10.1016/S2352-4642(23)00020-2. [Context Link 1] View abstract...
240. Son MBF, et al. Multisystem inflammatory syndrome in children - initial therapy and outcomes. *New England Journal of Medicine* 2021;385(1):23-34. DOI: 10.1056/NEJMoa2102605. [Context Link 1] View abstract...
241. McArdle AJ, et al. Treatment of multisystem inflammatory syndrome in children. *New England Journal of Medicine* 2021;385(1):11-22. DOI: 10.1056/NEJMoa2102968. [Context Link 1] View abstract...
242. Liu E, et al. Guidelines for the prevention and management of children and adolescents with COVID-19. *European Journal of Pediatrics* 2022;181(12):4019-4037. DOI: 10.1007/s00431-022-04615-4. [Context Link 1] View abstract...
243. Gilhus NE. Myasthenia gravis. *New England Journal of Medicine* 2016;375(26):2570-2581. DOI: 10.1056/NEJMra1602678. [Context Link 1, 2] View abstract...

244. Gajdos P, Chevret S, Toyka KV. Intravenous immunoglobulin for myasthenia gravis. *Cochrane Database of Systematic Reviews* 2012, Issue 12. Art. No.: CD002277. DOI: 10.1002/14651858.CD002277.pub4. [Context Link 1, 2] View abstract...
245. Sanders SL, Agwan S, Hassan M, Bont LJ, Venekamp RP. Immunoglobulin treatment for hospitalised infants and young children with respiratory syncytial virus infection. *Cochrane Database of Systematic Reviews* 2023, Issue 10. Art. No.: CD009417. DOI: 10.1002/14651858.CD009417.pub3. [Context Link 1, 2] View abstract...
246. Ebadi H, Barth D, Bril V. Safety of plasma exchange therapy in patients with myasthenia gravis. *Muscle and Nerve* 2013;47(4):510-514. DOI: 10.1002/mus.23626. [Context Link 1] View abstract...
247. Katzberg HD, Barnett C, Merkies IS, Bril V. Minimal clinically important difference in myasthenia gravis: outcomes from a randomized trial. *Muscle and Nerve* 2014;49(5):661-665. DOI: 10.1002/mus.23988. [Context Link 1] View abstract...
248. Bril V, et al. Randomized double-blind placebo-controlled trial of the corticosteroid-sparing effects of immunoglobulin in myasthenia gravis. *Neurology* 2023;100(7):e671-e682. DOI: 10.1212/WNL.0000000000201501. [Context Link 1] View abstract...
249. Hellmann MA, Mosberg-Galili R, Lotan I, Steiner I. Maintenance IVIg therapy in myasthenia gravis does not affect disease activity. *Journal of the Neurological Sciences* 2014;338(1-2):39-42. DOI: 10.1016/j.jns.2013.10.043. [Context Link 1] View abstract...
250. Skeie GO, et al. Guidelines for treatment of autoimmune neuromuscular transmission disorders. *European Journal of Neurology* 2010;17(7):893-902. DOI: 10.1111/j.1468-1331.2010.03019.x. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
251. de Alarcon PA, et al. Intravenous immunoglobulin with prednisone and risk-adapted chemotherapy for children with opsoclonus myoclonus ataxia syndrome associated with neuroblastoma (ANBL00P3): a randomised, open-label, phase 3 trial. *Lancet Child and Adolescent Health* 2018;2(1):25-34. DOI: 10.1016/S2352-4642(17)30130-X. [Context Link 1, 2, 3] View abstract...
252. Patel A, et al. Treatment and revaccination of children with paraneoplastic opsoclonus-myoclonus-ataxia syndrome and neuroblastoma: The Memorial Sloan Kettering experience. *Pediatric Blood and Cancer* 2020;67(8):Online. DOI: 10.1002/pbc.28319. [Context Link 1] View abstract...
253. Mitoma H, Manto M, Hampe CS. Immune-mediated cerebellar ataxias: practical guidelines and therapeutic challenges. *Current Neuropharmacology* 2019;17(1):33-58. DOI: 10.2174/1570159X16666180917105033. [Context Link 1] View abstract...
254. Gorman MP. Update on diagnosis, treatment, and prognosis in opsoclonus-myoclonus-ataxia syndrome. *Current Opinion in Pediatrics* 2010;22(6):745-50. DOI: 10.1097/MOP.0b013e32833fde3f. [Context Link 1] View abstract...
255. Hawkins J, Aster RH, Curtis BR. Post-transfusion purpura: current perspectives. *Journal of Blood Medicine* 2019;10:405-415. DOI: 10.2147/JBM.S189176. [Context Link 1, 2, 3] View abstract...
256. ACOG Practice Bulletin No. 207: thrombocytopenia in pregnancy. *Obstetrics & Gynecology* 2019;133(3):e181-e193. DOI: 10.1097/AOG.0000000000003100. [Context Link 1, 2] View abstract...
257. Neunert C, et al. The American Society of Hematology 2011 evidence-based practice guideline for immune thrombocytopenia. *Blood* 2011;117(16):4190-4207. DOI: 10.1182/blood-2010-08-302984. (Reaffirmed 2024 Jul) [Context Link 1, 2] View abstract...
258. Choi PY, et al. Consensus guidelines for the management of adult immune thrombocytopenia in Australia and New Zealand. *Medical Journal of Australia* 2022;216(1):43-52. DOI: 10.5694/mja2.51284. [Context Link 1] View abstract...
259. Octagam [immune globulin intravenous (human)] 5% liquid preparation. Physician Prescribing Information [Internet] Octapharma. 2022 Apr Accessed at: <https://octagamusa.com/octagam5/>. [created 2004; accessed 2024 Sep 30] [Context Link 1, 2]
260. Panzyga (immune globulin intravenous, human). Physician Prescribing Information [Internet] Octapharma USA Inc. 2021 Jan Accessed at: <https://www.octapharmausa.com/products/product-overview/panzyga/>. [created 2018; accessed 2024 Sep 30] [Context Link 1]
261. Asceniv (immune globulin intravenous, human - slra). Physician Prescribing Information [Internet] ADMA Biologics. 2022 Dec Accessed at: <https://www.asceniv.com/>. [created 2019; accessed 2024 Oct 03] [Context Link 1]
262. Cuvitru, immune globulin subcutaneous (human), 20% solution. Physician Prescribing Information [Internet] Takeda Pharmaceuticals USA. 2023 Mar Accessed at: <https://www.cuvitru.com/>. [created 2016; accessed 2024 Sep 24] [Context Link 1, 2]
263. Cutaquig (immune globulin subcutaneous (human) - hipp), 16.5% solution. Physician Prescribing Information [Internet] Octapharma USA Inc. 2021 Nov Accessed at: <https://www.cutaquigus.com/>. [created 2018; accessed 2024 Sep 24] [Context Link 1]
264. Xembify (immune globulin subcutaneous, human - klhw) 20% solution. Physician Prescribing Information [Internet] Grifols Therapeutics, Inc. 2024 Jul Accessed at: <https://www.xembify.com/>. [created 2019; accessed 2024 Sep 06] [Context Link 1]
265. Bivigam immune globulin intravenous (human) 10% injection, solution. Physician Prescribing Information [Internet] Biotest Pharmaceuticals Corporation. 2022 Dec Accessed at: <https://www.bivigam.com/>. [created 2012; accessed 2024 Oct 03] [Context Link 1]
266. Bonilla FA, et al. Practice parameter for the diagnosis and management of primary immunodeficiency. *Journal of Allergy and Clinical Immunology* 2015;136(5):1186-1205.e78. DOI: 10.1016/j.jaci.2015.04.049. [Context Link 1, 2, 3] View abstract...
267. Hoernes M, Seger R, Reichenbach J. Modern management of primary B-cell immunodeficiencies. *Pediatric Allergy and Immunology* 2011;22(8):758-769. DOI: 10.1111/j.1399-3038.2011.01236.x. [Context Link 1, 2] View abstract...
268. Knutsen AP, Leiva LE, Caruthers C, Rodrigues J, Sorensen RU. Streptococcus pneumoniae antibody titres in patients with primary antibody deficiency receiving intravenous immunoglobulin (IVIg) compared to subcutaneous immunoglobulin (SCIG). *Clinical and Experimental Immunology* 2015;182(1):51-56. DOI: 10.1111/cei.12665. [Context Link 1] View abstract...
269. Bonagura VR. Using intravenous immunoglobulin (IVIg) to treat patients with primary immune deficiency disease. *Journal of Clinical Immunology* 2013;33 Suppl 2:S90-S94. DOI: 10.1007/s10875-012-9838-1. [Context Link 1] View abstract...
270. Schroeder HW, Dougherty CJ. Review of intravenous immunoglobulin replacement therapy trials for primary humoral immunodeficiency patients. *Infection* 2012;40(6):601-611. DOI: 10.1007/s15010-012-0323-9. [Context Link 1] View abstract...
271. Freeman CM, Squire JD, Joshi AY. Immunoglobulin treatment for B-cell immunodeficiencies. *Journal of Immunological Methods* 2022;509:Online. DOI: 10.1016/j.jim.2022.113336. [Context Link 1] View abstract...

272. Orange JS, Grossman WJ, Navickis RJ, Wilkes MM. Impact of trough IgG on pneumonia incidence in primary immunodeficiency: A meta-analysis of clinical studies. *Clinical Immunology* 2010;137(1):21-30. DOI: 10.1016/j.clim.2010.06.012. [Context Link 1] View abstract...
273. Tuerlinckx D, et al. Pneumococcal antibody levels in children with PID receiving immunoglobulin. *Pediatrics* 2014;133(1):e154-e162. DOI: 10.1542/peds.2013-1155. [Context Link 1] View abstract...
274. Maguire T, et al. Home-Based Subcutaneous Infusion of Immunoglobulin for Primary and Secondary Immunodeficiencies. Ontario Health Technology Assessment Series [Internet] Health Quality Ontario Ministry of Health and Long-Term Care. 2017 Nov Accessed at: <https://www.hqontario.ca/>. [accessed 2024 Sep 25] [Context Link 1]
275. Gathmann B, et al. Clinical picture and treatment of 2212 patients with common variable immunodeficiency. *Journal of Allergy and Clinical Immunology* 2014;134(1):116-126. DOI: 10.1016/j.jaci.2013.12.1077. [Context Link 1] View abstract...
276. Lourdes LS, Daily KC. Common variable immunodeficiency syndrome in an adult. *Lancet* 2014;383(9920):926. DOI: 10.1016/S0140-6736(14)60415-3. [Context Link 1] View abstract...
277. Varadkar S, et al. Rasmussen's encephalitis: clinical features, pathobiology, and treatment advances. *Lancet Neurology* 2014;13(2):195-205. DOI: 10.1016/S1474-4422(13)70260-6. [Context Link 1, 2] View abstract...
278. Bien CG, et al. Rasmussen encephalitis: incidence and course under randomized therapy with tacrolimus or intravenous immunoglobulins. *Epilepsia* 2013;54(3):543-550. DOI: 10.1111/epi.12042. [Context Link 1, 2] View abstract...
279. Orsini A, et al. Rasmussen's encephalitis: From immune pathogenesis towards targeted-therapy. *Seizure* 2020;81:76-83. DOI: 10.1016/j.seizure.2020.07.023. [Context Link 1] View abstract...
280. Mockenhaupt M. Stevens-Johnson syndrome and toxic epidermal necrolysis: clinical patterns, diagnostic considerations, etiology, and therapeutic management. *Seminars in Cutaneous Medicine and Surgery* 2014;33(1):10-16. [Context Link 1] View abstract...
281. Schneider JA, Cohen PR. Stevens-Johnson syndrome and toxic epidermal necrolysis: a concise review with a comprehensive summary of therapeutic interventions emphasizing supportive measures. *Advances in Therapy* 2017;34(6):1235-1244. DOI: 10.1007/s12325-017-0530-y. [Context Link 1, 2] View abstract...
282. Tsai TY, et al. Treating toxic epidermal necrolysis with systemic immunomodulating therapies: A systematic review and network meta-analysis. *Journal of the American Academy of Dermatology* 2021;84(2):390-397. DOI: 10.1016/j.jaad.2020.08.122. [Context Link 1] View abstract...
283. Barron SJ, Del Vecchio MT, Aronoff SC. Intravenous immunoglobulin in the treatment of Stevens-Johnson syndrome and toxic epidermal necrolysis: a meta-analysis with meta-regression of observational studies. *International Journal of Dermatology* 2015;54(1):108-115. DOI: 10.1111/ijd.12423. [Context Link 1] View abstract...
284. McKeon A. Paraneoplastic and other autoimmune disorders of the central nervous system. *Neurohospitalist* 2013;3(2):53-64. DOI: 10.1177/1941874412453339. [Context Link 1] View abstract...
285. Yildirim-Toruner C, Diamond B. Current and novel therapeutics in the treatment of systemic lupus erythematosus. *Journal of Allergy and Clinical Immunology* 2011;127(2):303-314. DOI: 10.1016/j.jaci.2010.12.1087. [Context Link 1, 2] View abstract...
286. Bayry J, Negi VS, Kaveri SV. Intravenous immunoglobulin therapy in rheumatic diseases. *Nature Reviews. Rheumatology* 2011;7(6):349-359. DOI: 10.1038/nrrheum.2011.61. [Context Link 1] View abstract...
287. Gordon C, et al. The British Society for Rheumatology guideline for the management of systemic lupus erythematosus in adults. *Rheumatology* 2018;57(1):e1-e45. DOI: 10.1093/rheumatology/kex286. (Reaffirmed 2024 Sep) [Context Link 1] View abstract...
288. Sakthiswary R, D'Cruz D. Intravenous immunoglobulin in the therapeutic armamentarium of systemic lupus erythematosus: a systematic review and meta-analysis. *Medicine* 2014;93(16):e86. DOI: 10.1097/MD.0000000000000086. [Context Link 1] View abstract...
289. Camara I, et al. Treatment with intravenous immunoglobulins in systemic lupus erythematosus: a series of 52 patients from a single centre. *Clinical and Experimental Rheumatology* 2014;32(1):41-47. [Context Link 1] View abstract...
290. Fanouriakis A, et al. EULAR recommendations for the management of systemic lupus erythematosus: 2023 update. *Annals of the Rheumatic Diseases* 2024;83(1):15-29. DOI: 10.1136/ard-2023-224762. (Reaffirmed 2024 Jul) [Context Link 1] View abstract...
291. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2024 Oct 08] Accessed at: <https://www.ecfr.gov/>. [Context Link 1]
292. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2024 Sep 30 [accessed 2024 Oct 08] Accessed at: <https://www.ecfr.gov/>. [Context Link 1]
293. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 14 - Medical Devices Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2024 Sep 26] [Context Link 1]
294. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12425 [Internet] Centers for Medicare & Medicaid Services. 2023 Dec 21 Accessed at: <https://www.cms.gov/manuals/>. [accessed 2024 Feb 01] [Context Link 1]
295. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 16 - General Exclusions From Coverage Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2024 Jun 06] [Context Link 1]
296. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: <https://www.cms.gov/medicare-coverage-database/search.aspx?> Updated 2024 [accessed 2024 Sep 26] [Context Link 1]

Footnotes

[A] For acquired red cell aplasia, expert opinion suggests IVIG be administered as an intravenous infusion over 5 days.(83) Repeat infusions may be necessary.(84) [A in Context Link 1]

[B] For acute disseminated encephalomyelitis, expert opinion suggests IVIG may be administered intravenously for 4 to 5 days.(30) [B in Context Link 1]

[C] Catastrophic antiphospholipid syndrome (CAPS) is a clinical syndrome with 4 diagnostic criteria: (1) evidence of vessel occlusion in 3 or more organs, tissues, or systems; (2) development of manifestations simultaneously or within a short period (eg, 1 week); (3) biopsy confirmation of small vessel occlusion in at least one organ; (4) laboratory confirmation of antiphospholipid antibodies (ie, lupus anticoagulant or anti-cardiolipin antibody). The presence of all 4 criteria leads to a definite diagnosis of CAPS.(97) [C in Context Link 1]

[D] For autoimmune bullous diseases, expert opinion suggests intravenous administration of IVIG over 2 to 5 days approximately every 4 weeks until control is achieved, followed by gradual lengthening of dosing intervals by 2-week intervals.(76) IVIG should be discontinued if there is no response after 6 months.(76) [D in Context Link 1]

[E] For chronic inflammatory demyelinating polyneuropathy, for initial treatment or for relapse, IVIG is administered by intravenous infusion, with a second course of therapy approximately 3 weeks after the first if needed. For those requiring maintenance therapy, IVIG may be administered intravenously every 3 to 8 weeks as needed.(125)(126)(127)(128)(129) Alternatively, SCIG may be administered for maintenance therapy at intervals ranging from every 1 to 4 weeks starting 1 to 2 weeks after the last IVIG infusion; for patients whose symptoms worsen while on SCIG therapy, consideration should be given to reinitiating treatment with an IVIG-approved therapy and discontinuing subcutaneous administration.(130)(131)(132) [E in Context Link 1]

[F] For dermatomyositis refractory to immunosuppressive therapy, IVIG may be administered every 4 weeks.(139) An expert consensus guideline indicates that IVIG should be discontinued if no improvement occurs after up to 6 infusions. After 18 treatment cycles, a washout period is recommended.(76) [F in Context Link 1]

[G] For antenatal treatment of fetal-neonatal alloimmune thrombocytopenia, expert opinion suggests a weekly dose before delivery.(146) [G in Context Link 1]

[H] For hemolytic disease of the newborn, expert opinion suggests a single intravenous administration, with repeat administration in 12 hours if necessary.(167) [H in Context Link 1]

[I] Hemophagocytic syndrome is a rare but potentially fatal hyperinflammatory syndrome characterized by fever, hepatosplenomegaly, and cytopenias. It may be primary, with a genetic etiology, or secondary, associated with autoimmune disease, infections, malignancies, and transplant.(174) [I in Context Link 1]

[J] For prophylaxis of bacterial infection in HIV-positive children with hypogammaglobulinemia, a systematic review suggests IVIG be administered as an intravenous infusion every 2 to 4 weeks.(179) [J in Context Link 1]

[K] Immune checkpoint inhibitors work by enhancing the activity of cytotoxic T cells, thereby increasing their antitumor effects. This enhanced activation leads to inflammatory or autoimmune-like side effects that can affect all organs, including the skin (eg, bullous dermatosis, severe cutaneous reactions), nervous system (eg, demyelinating disease, encephalitis, myasthenia gravis), lungs (eg, pneumonitis), gastrointestinal tract (eg, diarrhea), and liver (eg, hepatitis).(190)(191) [K in Context Link 1]

[L] For prevention of or postexposure prophylaxis for hepatitis A, immune globulin is administered as a single intramuscular injection.(198) [L in Context Link 1]

[M] For postexposure prophylaxis for measles, immune globulin is administered as a single intramuscular injection or intravenous infusion.(199) [M in Context Link 1]

[N] For postexposure prophylaxis for rubella, immune globulin is administered as a single intramuscular injection.(200) [N in Context Link 1]

[O] For postexposure prophylaxis for varicella when varicella zoster-specific immune globulin is unavailable, IVIG is administered as a single intravenous infusion.(203) [O in Context Link 1]

[P] For Kawasaki disease, a specialty society guideline recommends that IVIG be administered intravenously as a single bolus infusion. In patients who are considered IVIG resistant (ie, with persistent or recrudescent fever at least 36 hours and less than 7 days after completion of the first IVIG infusion), a repeat IVIG infusion may be considered.(209) [P in Context Link 1]

[Q] For acute antibody-mediated rejection of a kidney transplant from either a deceased or live donor, expert opinion suggests IVIG be administered intravenously after each plasmapheresis treatment.(72) [Q in Context Link 1]

[R] For kidney transplant in patients with anti-HLA antibodies to living donor kidneys, expert opinion suggests IVIG be administered intravenously every month for 4 months to achieve a negative anti-human globulin complement-dependent cytotoxicity crossmatch prior

to transplant.(72) IVIG is less effective for desensitization of patients with high anti-HLA antibody titers greater than 1:16.(72) Treatment with either IVIG alone or plasmapheresis followed by IVIG is an acceptable regimen during the perioperative period.(72) [R in Context Link 1]

[S] For multifocal motor neuropathy, expert opinion suggests IVIG be administered by intravenous infusion over 2 to 5 days. If initial treatment with IVIG is effective, repeat IVIG should be guided by response. Typical maintenance treatment regimens vary and may be administered from every 2 to 4 weeks to every 1 to 2 months.(129)(228) [S in Context Link 1]

[T] The Centers for Disease Control and Prevention defines multisystem inflammatory syndrome in children as current or recent severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection in a patient younger than 21 years with fever greater than 100.4 degrees F (38 degrees C) for at least 24 hours, laboratory evidence of inflammation (eg, elevated C-reactive protein, erythrocyte sedimentation rate, fibrinogen, procalcitonin, D-dimer), and evidence of clinically severe illness requiring hospitalization with multiorgan system involvement (eg, cardiovascular, respiratory, renal, gastrointestinal, hematologic) without identified alternative cause.(233)(234) [T in Context Link 1]

[U] For multisystem inflammatory syndrome in children, an expert consensus guideline suggests that IVIG be administered by intravenous infusion over 1 to 2 days. A second dose of IVIG is not recommended in patients with refractory disease, given a higher risk of IVIG complications (eg, hemolytic anemia, volume overload) in this patient population.(234) [U in Context Link 1]

[V] For opsoclonus-myoclonus syndrome, a randomized trial in pediatric patients evaluated IVIG administered as an intravenous infusion over 2 days for initial therapy, followed by infusion over 1 day every 3 to 4 weeks for 5 infusions, and thereafter over 1 day every 6 to 8 weeks.(251) Treatment duration should be individualized based on symptoms.(252) [V in Context Link 1]

[W] For post-transfusion purpura, expert opinion suggests administering IVIG as a daily intravenous dose for 1 to 10 days.(255) [W in Context Link 1]

[X] For primary immunodeficiency, IVIG is administered as an intravenous infusion every 3 to 4 weeks.(11)(125)(129)(205)(259)(260)(261) SCIG is administered for primary immunodeficiency at regular intervals from daily up to every 4 weeks.(125)(126)(129)(130)(132)(183)(262)(263)(264) [X in Context Link 1]

[Y] For toxic epidermal necrolysis, expert opinion suggests IVIG be administered in a single course as soon as possible after diagnosis.(76) [Y in Context Link 1]

Codes

HCPCS: J1459, J1551, J1552, J1554, J1555, J1556, J1557, J1558, J1559, J1561, J1566, J1568, J1569, J1572, J1575, J1576, J1599, Q2052, S9338, S9563 [Hide]

MCG Health
Ambulatory Care 29th Edition
Copyright © 2025 MCG Health, LLC
All Rights Reserved

Last Update: 6/13/2025 3:23:34 AM
Build Number: 29.1.20250613001137.022065